Degree Project in Integrated Product Design

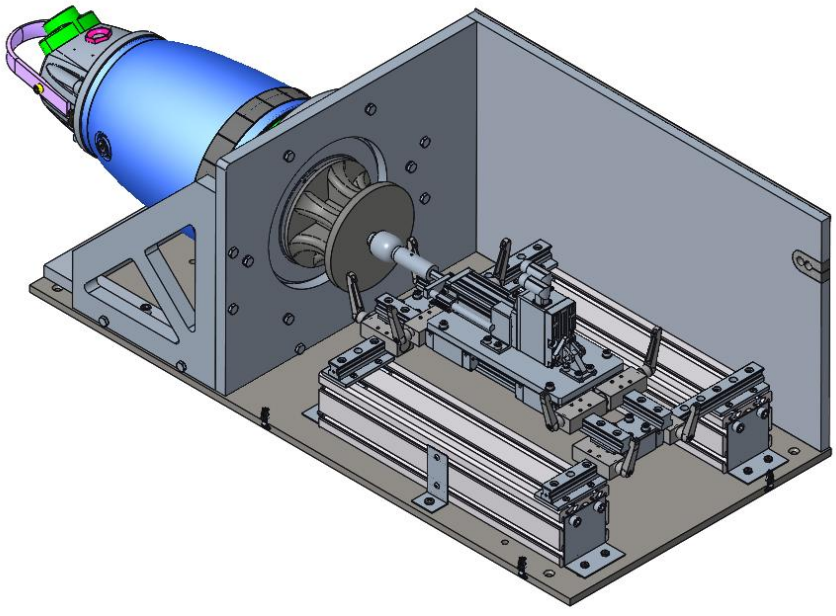
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Development of a conceptual test rig for an adaptive mechanism in wastewater pumps

Design and evaluation of a test setup for impellers with adaptive mechanisms at Xylem

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Master's Thesis

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Abstract

Wastewater pumps function in harsh environments filled with debris and larger objects such as gravel and wet wipes. Fibrous materials prevent the pump impeller from rotating, causing clogging, leading to increased operational and maintenance costs. To reduce clogging, Xylem has developed an adaptive mechanism within the impeller that allows larger flow to pass through the pump. The purpose of this thesis was to develop a conceptual test rig capable of recreating the damages inflicted on the adaptive mechanism, but within an accelerated timeframe.

Currently at Xylem a method has been developed to test the mechanism within an existing pump. However, this method is labour intensive, time consuming and very loud. One of the department's focus areas is the development of the adaptive mechanism, leaving limited time to develop a separate test rig. Their current method does not identify which specific motion on the impeller causes what damage in the mechanism.

First, the damage caused by long time use of the impeller was examined. Then forces acting on the mechanism were identified and tied together with the relevant damage. Different test rigs at the company were explored, with the aim to find inspiration. The concept development followed an iterative process, where several concept generation methods were used. The multiple concepts were then compared to the requirement specification to select one of the concepts. The selected test rig concept was further developed through material and component selection, three-dimensional modelling, simplified laboratory testing and solid mechanics calculations.

The final concept is called a repetitive movement test rig allowing for two different accelerated damage tests. One recreates the linear collisions caused by objects passing through the pump, and the other recreates misalignment of the impeller as these objects passes through. In the linear test a centrally aligned linear actuator applies axial loads on a simplified version of the impeller. During the misalignment test the linear actuator will be moved along linear guide rails to apply the force near the outer diameter of the impeller. After either of these tests has been conducted, the adaptive mechanism will be removed from the simplified impeller, and any damages will be studied and noted. Together these tests allow the user to identify what motions contribute to specific damage within the adaptive mechanism. Building on this conceptual design, the continued construction of a less labour intensive test rig can be done by the development department at Xylem.

Keywords

Adaptive mechanism, Wastewater pump, Test rig, Accelerated test, Automatic testing, Linear test, Misalignment test, Combined movement

Sammanfattning

Avloppspumpar opererar i en utmanande miljö, där vattnet innehåller objekt så som grus och hygienartiklar. Objekt gjorda av fiber förhindrar pumphjulet från att rotera och orsakar igensättning. Igensättning leder till ökade drifts och underhållningskostnader. För att minska igensättningen har Xylem utvecklat en adaptiv mekanism inuti pumphjulet som tillåter större objekt att passera. Syftet med denna rapport var att utveckla en konceptuell testrigg kapabel att återskapa de skador som uppstår på den adaptiva mekanismen under operation, men under en accelererad tid. Utvecklingsavdelningen har skapat en metod för att testa mekanismen i befintliga testriggar på företaget. Denna metod är dock arbetsintensiv, tidskrävande och högljudd. Teamet fokuserar på att utveckla den adaptiva mekanismen och därför finns det begränsad tid för de att utveckla en separat testrigg. Xylems nuvarande metod ger ej möjligheten att identifiera den specifik rörelsen på impellern som orsakar skador i mekanismen.

Först undersöktes skadorna orsakade efter långvarig användning av mekanismen. Därefter identifierades krafter som verkar på mekanismen och relaterades till de skador de orsakat. Olika testriggar på företaget studerades, i syfte att få inspiration. Konceptutvecklingen följde en iterativ process där flera konceptgenereringsmetoder användes. De olika koncepten jämfördes med kravspecifikation för att möjliggöra val av ett koncept. Det valda testriggkonceptet vidareutvecklades och material valdes. Därefter testades konceptet i ett förenklat test i laboratoriet, en CAD modell skapades och hållbarheten på testriggen beräknades med hållfasthetsberäkningar.

Det slutliga konceptet möjliggör två olika accelererade skadetester. Det ena återskapar de linjära kollisionerna som orsakas av objekt som passerar genom impellern, den andra återskapar snedställningen av pumphjulet när dessa objekt applicerar en momentkraft. I det linjära testet applicerar en linjärmotor den axiella belastningen på en förenklad version av impellern. Under snedställningstestet kommer den linjära motorn förflyttas längs linjära styrskenor för att applicera kraften nära pumphjulets ytterdiameter. När testet har utförts kommer den adaptiva mekanismen att tas bort från testhjulet, och eventuella skador kommer att studeras och noteras. Tillsammans gör dessa tester det möjligt för användaren att identifiera vilka rörelser som bidrar till specifika skador inom den adaptiva mekanismen. Med utgångspunkt i denna konceptuella design kan den fortsatta konstruktionen av en bättre presterande testrigg utföras av utvecklingsavdelningen på Xylem.

Nyckelord

Adaptiv mekanism, Avloppspump, Testrigg, Accelererat test, Automatisk testning, Linjärt test, Snedställningstest, Kombinerad rörelse

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Stockholm, June 2026
Jes & Jos

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List Of Abbreviations

RPM - Revolution Per Minute

LT - Low Head Trim

MT - Medium Head Trim

HT - High Head Trim

RMTR - Repetitive Movement Test Rig

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1 Introduction

The following chapter describes the problem, research methodology and delimitation of the project.

1.1 Background

The transportation of large volumes of liquid over long distances is an essential function in modern infrastructure. Pumps are used in industries and municipalities to pump liquids from one location to another. One use of the pump is to move wastewater through the pipes of households and commercial buildings to cleaning facilities. In order to move the water, the pump converts electrical energy to mechanical energy by rotating an impeller. This action generates centrifugal forces that accelerates the fluid out from the centre of the impeller. This creates a lower pressure under the impeller, forcing more of the liquid through the wastewater pump. This device work in harsh environments, having to withstand sewage water containing waste, grease, chemicals and other debris. Moreover, the pump needs to manage larger objects and debris commonly found in the sewage systems, such as wet wipes or gravel. Since the pump rotates at a high speed and the impeller vanes must maintain close contact with hydraulic chamber to create low pressure, objects in the sewage water can become lodged in the pump. If the object is solid, it can damage the impeller or get wedged in between the impeller and the edges, causing the impeller to stop rotating. Another issue is fabrics and stringy objects that get twisted in the impeller and cause a clog. Clogging occurs when fibrous or string-like materials become entangled in the impeller, preventing it from rotating properly. This is a regular problem in wastewater pumps, and the consequences are serious involves high costs and maintenance work, since clogged pumps leads to an increase in energy consumption for the user (Xylem, 2017).

Xylem is a global leader in water pumps; the company has a range of products pumping liquids from one location to another. The pumps vary greatly in size and functionality, since they are intended for different situations. One of the challenges Xylem faces is clogging in the wastewater pumps, which limits their effectiveness. To avoid clogging, Xylem developed a mechanism allowing a large amount of water flow for a short period of time. It reduces the risk of clogging and maintains a high efficiency and is called separation in time. The mechanism is located inside the impeller and allows it to move in the axial orientation. With this axial movement a leeway is created, enabling waste such as wipes, hygiene articles and dishcloths through. By allowing the impeller to move, the objects do not get lodged in it as they are forced through the pump.

1.2 Problem

The issue today is that when the adaptive mechanism deforms it is challenging to analyse which forces are causing damage. There are three major issues that the impeller can be exposed to: misalignment, leakage and mechanical axial collision. Xylem today has several test rigs for the wastewater pumps, testing different conditions they will be exposed to. However, Xylem lacks a test rig specifically targeting the adaptive mechanism. Therefore, it is challenging to determine which forces are responsible for the different types of damage occurring within the mechanism. This information is needed when developing new prototypes to give help with design choices. Today, one test is performed to target the adaptive mechanism, but the process is labour intensive and time consuming. The mechanism can be evaluated after operating under real conditions, but an impeller typically has a service life of around decade, making it impractical to wait for damage to develop naturally (Xylem, 2017).

Because of these challenges this thesis will focus on developing a test rig simulating the damages caused within the adaptive mechanism, in an accelerated time frame. Therefore, the following research questions have been formulated:

1. Which parameters are most relevant to identify damage in the adaptive mechanism?
2. What motions should the test rig simulate to reproduce the damage in the mechanism?
3. How can the test rig be designed to minimise manual labour during the test?

1.3 Purpose

The purpose of this thesis is to describe the development of a conceptual test rig, for investigating damage development in the adaptive mechanism of a wastewater pump. The test rig is intended to simulate repeated loads until failure or damage of the adaptive mechanism occurs. The number of impulses to failure and resulting mechanical wear will be evaluated to compare different mechanism prototypes.

1.4 Goals

The sub-goals given from the commissioner for this project were:

- Definition of requirements and measurement parameters based on the impeller operating environment
- Generation and comparison of multiple rig concepts
- Definition of functional and safety requirement specification for the rig
- Definition of the test methodology, including data collection and data analysis procedures
- Evaluation of dimensions, material selection and strength calculations
- Creation of test rig component list
- Creation of a digital model of the test rig
- Creation of drawing of the test rig
- Evaluation of the rig's performance and provide recommendations for further development

1.5 Research Methodology

The project follows the Double Diamond design framework, which consists of challenge, empathising, defining, idea generating, prototyping and testing (Openlab, nd). The chosen framework was applied due to the development-oriented nature of the project. The initial project objective did not specify the exact testing purposes therefore; the project required an iterative process to allow continuous refinements. The project was divided into several phases, beginning with project planning. During this phase the scope and timeline was defined. Following this, the information gathering phase helped clarify the context of the issue. Informal and semi-structured interviews were conducted to gain practical insights on how test rigs are currently designed and operated. The information from the interviews was complemented with white and grey literature, providing a theoretical foundation on general test methodologies and allowing inspiration to be drawn from testing methods in other industries.

Based on this, reliability tools including, fault tree analysis was applied to understand what motions the test rig should simulate. Moreover, a requirement specification was established, defining the functional requirements and parameters to be measured in the test rig. Concept generation was then carried out using concept generation methods, such as Crazy 8. This resulted in multiple concepts, that were evaluated and

narrowed down to a final design. Validation was performed through strength calculations and simplified prototype testing to ensure that the design met the defined requirements. The final phase involved the creation of 3D models, drawings, and component selection for the test rig.

1.6 Delimitations

This project only examined the two models of Concertor pumps 6020 and 6030. The focus was solely on how to test the adaptive function and not explore how to improve it. The project examined how the impeller mechanism has broken before, only to determine what motions and parameters the test rig should target. The thesis does not include building the actual physical functioning test rig or testing it in practice. The concept does not include interface, or how the equipment for parameters will be assembled. Because of confidentiality some of the components in the adaptive mechanism will be simplified or omitted. The final drawings will also be omitted for confidentiality reasons. The number of participants and their names have been excluded because of ethical guidelines.

1.7 Structure Of The Thesis

Chapter one introduces the background and context of the thesis. The second chapter shows the frame of reference and a general description of wastewater pumps and the adaptive mechanism, which leads to chapter three the literature study. Chapter four shows the predesign, which influenced how the test rig would be developed. The next chapter explains the concept generations, which leads to the chosen concept. Chapter six is development of the concept The concept is further developed, with strength calculations and final design. The thesis ends with a discussion and future work.

2 Frame Of Reference

This chapter includes the frame of reference needed to execute the project, such as an explanation of general wastewater pumps and the adaptive mechanism. Description of other test rigs at Xylem is also included.

2.1 Wastewater Pump

Most wastewater pumps used in partially submerged conditions are built as close coupled units. The general part of a pump is the motor, pump chamber, seal system, suction inlet, discharge outlet, bearings, shaft and cooling system. The motor is mounted on a drive shaft which is connected to the impeller. The upper part of the wastewater pump is the drive unit, while the lower part is the hydraulic unit. The pump chamber is where the impeller is located and since the impeller is fully submerged in liquid it must be sealed from the electrical motor (Xylem, 2018). The entire wastewater pump can be seen in Figure 1.

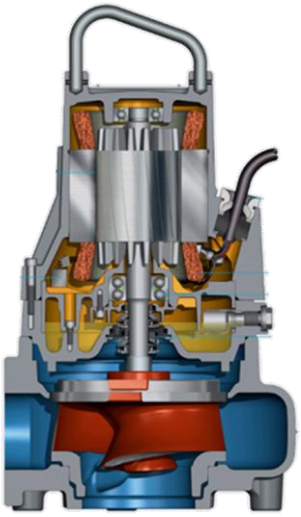


Figure 1 Cross section of a low-capacity Xylem pump

Since the environment the pump operates in is harsh, the impeller must shred and crush foreign objects or material when it goes through the pump (Nguyen et al., 2023). The stress of dealing with these materials leads to damage and abrasion to the impeller and pump. Moreover, another common problem with wastewater pumps is clogging, this occurs mostly due to fibrous materials such as wipes and hygiene articles that accumulate in the pump.

The impeller consists of a shroud, which is a plate where the vanes are positioned (Donahue, 2024). Between the vane and the shroud, the hub functions as a link between the parts; see Figure 2. The process of the pumping begins with the motor rotating the impeller with a drive shaft. The impeller functions as a connection between the motor and the fluid movement (Donahue, 2024). It uses centrifugal force to draw wastewater into the suction inlet vertically and then through the impeller and outwards to the discharge outlet. The rotating impeller translates energy to the fluid moving it uphill and outwards. This energy creates low-pressure in the middle of the impeller, also named the eye (Wilo, 2025). The foundational geometry principle for impellers is that the flow that the pumps generate is affected by the height of the impeller vane (Donahue, 2024). A tall impeller vane height will generate more flow than a shorter impeller vane. Another geometry principle for impellers is that the outer diameter of the vanes on the impeller varies the pressure the pumps generate (Donahue, 2024). A larger outer diameter will generate higher pressure and likewise with a smaller outer diameter.

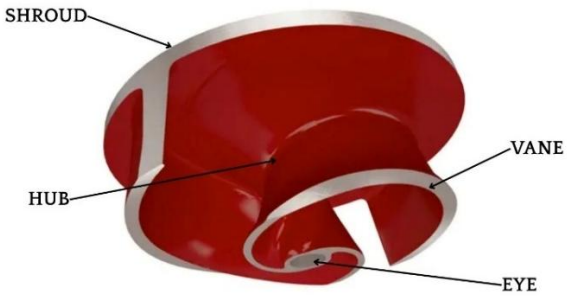


Figure 2 The different parts of the impeller

The adaptive mechanism consists of several components, due to confidentiality a similar version of the adaptive mechanism that have been studied in this project, which can be seen in Figure 3. The following components are used for the mechanism; 1. Inlet, 2. Outlet, 3. Impeller, 4. Compression spring, 5. Sleeve, 6. Pumphouse, 7. Guide-ring, 8. Insert, 9. Seal.

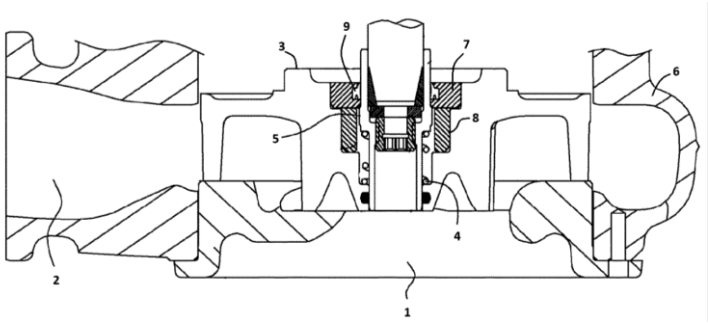


Figure 3 The different components of the adaptive mechanism

The adaptive mechanism contains a sleeve connected to the drive shaft; the sleeve translates the rotating motion from the shaft to the impeller through a spline coupling. The adaptive mechanism can be re-assembled if it is taken apart. When the axial force from an object is larger than the downwards force of the water the impeller moves upwards to the proximal position. This proximal position creates a leeway that can be seen in Figure 4. When the object has past, the impeller is forced down to its original position, the distal rest position by the force of the water. The impeller can move axially because the spline teeth on the sleeve is shorter than those on the insert.

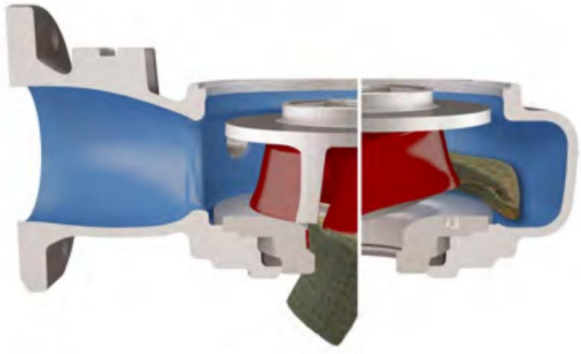


Figure 4 Adaptive impeller, distal rest to the left and proximal position to the right

The material for the impellers in the wastewater pumps varies between three common ones; grey iron, stainless steel and hard iron (Xylem, 2013). The chosen material depends on the environment of the pump, specifically the abrasive particles and the number of chloride ions. This is due to the materials having different resistance to, for instance corrosion and wear.

The thesis will only explore the wastewater pump of the model Concertor, which has two models 6020 and 6030; seen in Figure 5. For each of these models there are Low Head Trim (LT), Medium Head Trim (MT) and High Head Trim (HT) version of the impeller wheel. These pump systems have built-in intelligence, being able to clean the impeller when the system is clogged or working at a lower efficiency. It does this by stopping and reversing the rotational direction of the impeller when objects become lodged or the power consumption increases. The system saves energy and lowers the need for unintended services to unclog the pump (Xylem, 2024).

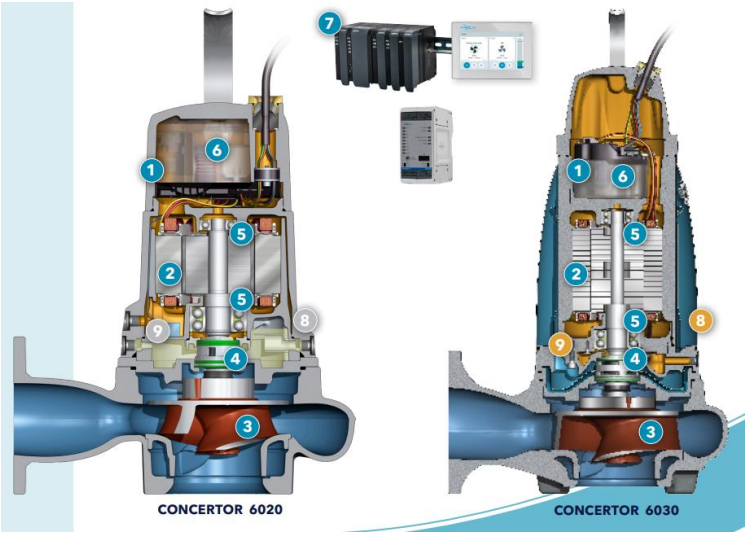


Figure 5 Cross section of Concertor 6020 and 6030

2.2 Forces On Impeller In Real Conditions

Due to the harsh and unpredictable environment of the wastewater pumps, there are several forces which impact the components in the impeller. From a semi-structured interview with a hydraulics engineer it was understood that the static radial force is caused by the flow of the water. Moreover, the impeller is also exposed to an axial force due to the pressure difference in the pump chamber. Other damage caused by the hydraulic forces is vibrations and leakage. The hydraulic forces can be seen in Figure 6.

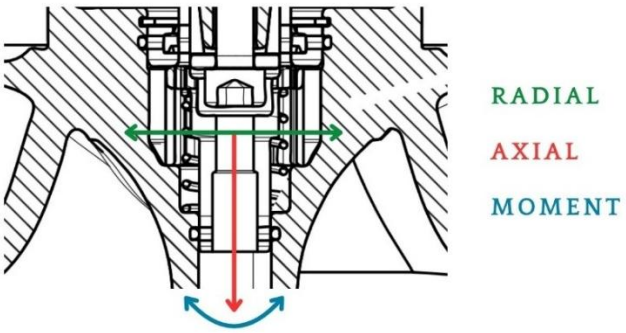


Figure 6 Hydraulic forces applied on the adaptive mechanism

The objects that get sucked into the impeller create their own forces, but it reacts in the opposite direction compared to the hydraulic forces. This force from the object forces the impeller to move to the proximal position. Additionally, the objects create a moment force on the impeller since they applied their forces on the vanes.

2.3 Related Mechanisms

A deeper understanding was achieved through the analysis of existing mechanisms with similar motion principles as the impeller, when objects pass through. One of the main challenges in the project was identifying practical solutions and existing mechanisms capable of combining both rotational and linear motion within the same mechanical body. Since this combination of movement was a desired feature for the proposed test rig, studying mechanisms with similar kinematic behaviour provided valuable design insights and served as an inspiration for the development process.

One mechanism in other products that have a similar movement as the adaptive mechanism is the computer numerical control (CNC) spindle head. The reason being that it has a high rotational speed and allows for axial movement (Greatcncmachine, 2024). The function of the spindle in the CNC machine is to hold and rotates the workpiece or cutting tool at a high speed. The rotational speed varies up to 60000 revolution per minute (RPM). This demonstrated how high-speed rotation can be combined with controlled axial positioning with a compact mechanical system.

Furthermore, another machine that combines both linear and rotational motion is a screw capping machine. The function of the machine is to screw caps onto bottles with the correct torque. The system picks a cap from a higher level, while rotating, and lowers the cap onto a bottle and screws it with a relevant torque. The specific method is called chuck capping and includes the rotating chuck that holds the cap and does the rotational and linear movement with it. A similar machine that executes the combined movement is screwdriving machines that use a robotic arm to screw objects into place. There are electric servomotors that rotate the screws with controlled torque and angle. The screwdriver moves in a linear motion down tightening the screws and then retracts.

These mechanisms collectively highlighted different approaches of integrating rotation and linear motion within a single system, either through spindle-based designs or guided linear actuator. This informed the conceptual development of the test rig by demonstrating feasible strategies for achieving combined motions while maintaining control over force, speed and positioning.

2.4 Other Test Rigs At Xylem

In order to gain inspiration for the design of the test rig in this project, the functions of the current test rigs at Xylem were studied. The laboratory has several test rigs and four of these were explored. The first one is the clog test rig (CTR), which is used to test the clogging capacity of the different wastewater pumps. The CTR is a large steel water tank filled with water, the pump is then lowered and secured in the rig and connected to pipes to pump water through. Water is continuously pumped, in a closed loop and fabrics are repetitively added to test the pumps anti-clogging abilities. The fabrics are added manually by the user throwing in the fabric during operation, thus being labour-intensive and time consuming. There are several test methods made in the CTR allowing the engineers to analyse pumps in different working conditions. Three different types of test methods were observed in the CTR, leading to an understanding of the motions acting upon the adaptive mechanism. Figure 7 illustrates the motion of fabrics passing through the impeller inlet, rotating with the impeller, and being discharged through the outlet.

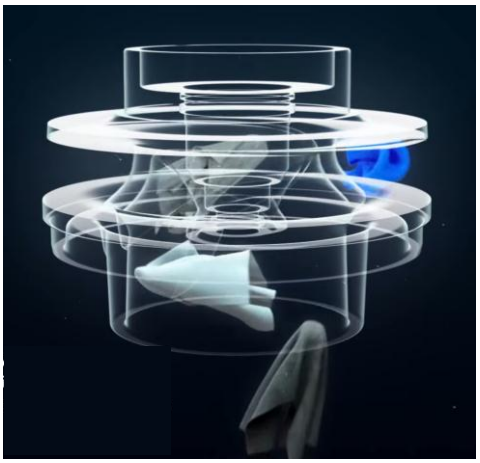


Figure 7 Picture of fabrics entering the impeller in a pump

The second test rig explored was for seals in wastewater pumps. The test was examined during operation several times to learn how it was prepared before a test and how data was gathered during the test. The test rig has a simplified version of the shaft, seal house and impeller unit. There are four different machines with each shaft dimension. In order to detect water leakage, a transparent oil-filled container is positioned on top of the assembly. The test runs for one month independently with the shaft rotating and water creating pressure on the seal. The motors efficiency, water temperature and water pressure are the parameters possible to adjust. The test rig is highly automatic and there is no need for monitoring or changing of the parameters during operation. This test rig does not intentionally have an accelerated time frame, however the combination of high pressure and continuous, uninterrupted operation results in greater damage, compared to the seals in the operation field. During one of the demonstrations of the seal test rig, one design engineer expressed a wish to see the seal at the beginning and end of the test. Currently, it is fully encapsulated in the metallic housing and can only be seen when disassembled. During one testing, however, an issue arose with the seal test rig. It was installed incorrectly which led to the rig being overheated and leaking water. This incident led to two important insight that, first that trouble shooting should be available without disassembling everything, and second that the temperature of the environment of the component is important. From the demonstration it was also observed that the test rig contained some sharp metal edges, which presented a safety hazard for the user.

The third test rig studied was the cable test rig, where the interviewee simultaneously demonstrated the operation of the test rig during the interview. The test is a simulation of the dynamic load the cables are exposed to when they are used in pumpstations. Because of this, one of the ends of the cable is fixed in the rig, while the other end is rotation at a chosen frequency. The rotational movement is symmetrical, although that is not a reflection of reality, as water causes the cables to move in an irregular and unpredictable manner. The test rig can evaluate up to ten cables simultaneously. The user may select both the operating frequency and the cable length, since these parameters affect the resulting cable quality. The test runs automatically once the user starts the test rig, causing the cables to rotate without requiring further user interaction. In addition, the test cycle is defined by specifying the number of hours the system will operate. When the test cycle was finished, the cables were detached from the rig, and the individual wires were visually and physically analysed and counted to record the number of failures. This procedure was repeated for each remaining cable. Conducting the test on several cables enabled a comparative assessment of which components experiences damage. The interviewee expressed that it is highly impossible to translate the number of hours the equipment runs in the rig to the number of hours the cable will survive in practice. It was understood that one possibility, for the test rig in this thesis, would be to allow for several impellers to

be tested at the same time. This is to compare the different impellers to each other to allow a benchmarking on how long the components can last as well as its degree of deformation.

Finally, a test rig used to evaluate pump efficiency was examined through both an interview and a demonstration. The test rig consists of two motors: one is the pump's original motor, while the second simulates the hydraulic forces acting on the drive shaft. The two shafts from each motor are connected by a jaw coupling, allowing the second motor to apply resistance to the system. Cooling is provided either through internal cooling systems or by directing compressed air into the surrounding casing. The pump-motor can be repositioned along grooves, further away from the second motor, to allow for easy replacement of the drive shaft. Torque and rotational speed are measured using a sensor placed between the shafts, while additional sensors record temperature, current, torque and power throughout the test. All data was logged continuously and made available online in real time. Similar to the two previous test rigs, this test rig also operated automatically, as the user only needed to start the system for the measurements to be recorded. The entire test rig was assembled on a trolley to allow moving the rig between different departments in the laboratory. For operator safety, a plexiglass barrier mounted on a sliding frame covers the rotating shafts during operation. Observing the test rig provided a deeper understanding of how parameters could be logged into a system. Moreover, it also influenced the physical structure of the test rig, with two motors at each end.

3 Literature Study

The following chapter is a summary of the research papers examined to gain a deeper understanding of the issue.

3.1 Accelerated Test

In order to determine how the mechanism fail the degradation of the components needs to be recreated in a shorter amount of time than in real conditions. It is important that the failures caused by the test with extremely increased forces are equal to the failures during real operation. The expected life of the components in real operation cannot be estimated from the accelerated stress test. The most important step for accelerated tests is deciding what stresses to apply. Early failure mechanisms can be identified by examining field failure data. Another important part of testing is detecting failure. Soft failures are the most difficult to catch as these change the conditions in the system and occur while the operation continues. The only way to identify these are with continues monitoring. The failure must also be connected to the stress that has been applied to the system, and therefore those must be monitored as well. In order to use the information from the tests a record should be kept, building on past experiences and recognising patterns (Thermotron Industries, 1998).

3.2 Test Rigs For Other Components

Test rigs for similar components and failure mechanisms were analysed to gain inspiration for the development process, relevant loading conditions, measured parameters and sensor selection. Particular focus was placed on components included in the adaptive mechanism, such as spring and sealing systems. The reason being, the purpose of the thesis was to accelerate wear and replicate damage in the adaptive mechanism components.

One of the components in the adaptive impeller is a spring, therefore test rigs for a spring was reviewed. Noskievic and Lara De León (2021) developed a rig to test the stiffness of springs. Spring stiffness was tested by slowly compressing a spring using a hydraulic actuator and measuring with sensors. The testing cycle began with placing the spring into a holder and entering the required compression. The input was used for the machine to signal the needed piston position. Thereupon, the data was logged and translated with MATLAB to create graphs of the springs change of stiffness, when it was compressed. This approach demonstrated how controlled compression with sensor feedback could be used to characterise component behaviour under loading. This was used in the selection of parameters and how it would work for the test rig in the thesis.

Another article was reviewed about a test rig trying to identify the potential failure cause of the aviation hydraulic rotary joint. This component shares similar features to the impeller mechanism, as it rotates with a shaft and contains seals to not get fluids in the system. To develop test equipment for the component, surfaces were first studied and then the fitting equipment was analysed with finite element methods (Zhang et al., 2023). The study highlighted the importance of analysing sealing interface and rotational assemblies when investigating failure mechanism. This was directly relevant for identifying critical wear points in the adaptive mechanism.

Klausen et al. (2017) developed an accelerated test rig for bearings. The accelerated test happened by increasing the load, but at a lower frequency. The authors test rig was performing heavy radial and axial loads on the bearings, at a low speed. The rotation per minute chosen for the test rig was 20, while the radial and axial loads were approximately 10 kN. Klausen et al. (2017) used several sensors for the test rig such as vibrator accelerometer, temperature sensor, acoustic emission sensor, shock pulse and proximity sensor.

This type of accelerated testing showed how increasing load while maintaining lower speeds could be used to induce controlled wear. This influenced the understanding of how damage could be replicated in a laboratory environment.

3.3 Friction Induced Heating

From the observation of the seal test rig, it was evident that the temperature of the environment has an impact on test objects. A similar consideration for the developed test rig would be how metal interacts with other metal surfaces, since several components in the adaptive mechanisms are made of metal. When two metal surfaces are in contact and move relative to each other at high speed, they create friction and heat. Consequences could be excessive wear and material degradation. If one metal surface is held stationary and a rotating metal surface is brought into contact, this could cause friction welding. For this phenomenon to occur pressure is also needed between the surfaces (Ahmad Mir et al., 2022). In systems where welding is not desired, excessive heat generation can be avoided with lubrication or a cooling system by reducing the friction.

3.4 Components Behaviour During Applied Load

One of the components in the adaptive mechanism is a spring. Therefore, it has been examined how springs works under certain loads, to understand potential failure modes relevant to the test rig in this thesis. The function of a spring is to manoeuvre external loads by absorbing shocks, minimise vibrations of a system and maintain a tension (Yang et al., 2014). Under a certain load the spring can break, and Yang et al. (2014) hypothesis was that this happens when the coils of the spring abruptly get into contact with each other. Even though springs in many systems do not necessarily reach its solid length. The solid length is the instance when the spring cannot compress any further, since there is no gap left for the coils to move. In this state the spring cannot twist, flex, bend or absorb any energy (Yang et al., 2014). Yang et al. (2014) Found that under sudden load, maximum shear stress it reached at the top of the spring when it reaches its solid height.

Spline coupling often works as a mechanical connection between a shaft and a hub. There are different types of spline teeth, one of them being straight-sided teeth (Billy, n.d.). Zhang et al. (2023) created a test with an assembly including spline coupling, drive shaft and bearings. According to Zhang et al. (2023) one of the troubles with spline couplings was misalignments since it resulted in accelerating the wear and uneven load distribution. Moreover, the spline coupling could be affected by several factors such as temperature, system vibration and working forces. There were two types of misalignments, static respectively dynamic (Zhang et al., 2023). Static misalignment was caused by inaccuracy of manufacturing and installation. External and internal spline displacement, caused by the vibration of the operating machinery was known as dynamic misalignment. Static and dynamic misalignment included parallel and angular misalignment. For static, parallel misalignment occurred when two shafts were approximately parallel but offset sideways. Static, angular misalignment happened when the shafts were at a slight angle in relation to each other. Dynamic, parallel misalignment was caused by transverse (side-to-side) vibration. While angular misalignment occurred by twisting or tilting motion. Zhang et al. (2023) discovered that single tooth fracture of the spline coupling was not likely due to the static angular misalignment. Moreover, increased static parallel misalignment caused for further uneven load distribution.

Pedersen (2011) executed an experiment with straight-sided splines that was exposed to pure torsional load. The findings were that there were high stress concentrations at the root of the tooth when torque was transmitted. Furthermore, the stress distribution of the circumferences was uneven, since it had peaks at specific areas which was at the internal fillet of each tooth. Pedersen (2011) explained that small root radii

could result in higher stress levels. The heightened stress levels would result in accelerated wear and spline tooth markings in components.

These findings indicate that the adaptive mechanism is highly sensitive to combined loading conditions, including compression and misalignment. This implies that the test rig must be capable of replicating the load cases in a controlled manner. By doing this, the test rig could properly accelerate wear and reproduce realistic failure modes in springs and spline-based connections.

4 Pre Design

The following chapter explains the process before generating concepts for the test rig. The test rig is intended to help the research and development (R&D) department identify weaknesses in the adaptive mechanism prototype design. The chapter also includes requirement specification and functional analysis of the rig.

4.1 Analysing Existing Damaged Impellers

The first step of the development process was to analyse existing impellers to understand what the test rig needed to test. This type of failure is uncommon; however, twelve impellers with failed adaptive mechanisms were analysed and compared with adaptive mechanisms that had been exposed to lower levels of wear. The breakage within the adaptive mechanism varied, although it was evident that the components were similarly damaged. An observation was that some of the failures within the mechanism might have been secondary failures, caused by another component initially failing. Abrasion, plastic deformation and scuffing marks were some of the identified damages. Other damage was a deep circumferential groove created by the spring. Some of these issues were tied to the continued collisions between components caused by the axial movement. With this information two reliability tools were conducted. Fault tree analysis (FTA) respectively failure mode and effects analysis (FMEA). The Failure tree analysis began by mapping which components had visible damage, thereafter, the potential causes for these were considered. Another potential cause was the components colliding with each other, because of the axial force pushing the impeller upwards to the proximal position. Moreover, a potential cause for damage is the impeller rotating continuously throughout the movement between proximal to distal rest position. The components were listed with the failure mode, cause and effect. The severity, occurrence, ability to detect and risk priority number (RPN) was decided by estimation. From the FMEA it was evident that the components with spline coupling had the highest RPN. The information gathered from these two tools was part of the decision on what motions and forces the test rig would simulate.

4.2 Understanding User Needs

A semi-structured interview was conducted to gain deeper understanding of the needs of the rig from the user's perspective. It was noted that the duration of the test performed by the test rig determines whether multiple components should be tested simultaneously. If the testing process is time consuming it will be more efficient to test several components in the adaptive mechanism at once, whereas shorter test times allow components to be tested individually. Moreover, the test rig would be used in periods, since it all depends on the number of prototypes and the need for improving the existing components. Another finding was the weight of the impeller, that it could be handled manually without any additional equipment, such as an overhead crane.

4.3 Requirement Specification

Objectives and constraints for the test rig were defined based on insights gained from the analysis of existing damaged impellers. These constraints combined with collected information, were used to formulate a requirement specification. Among the requirements were the need for the system to support various adaptive mechanisms, apply repetitive axial forces and apply rotational force. The requirement specification was developed during the entirety of the project and included functional requirements, size constraints, safety and regulatory requirements (see Appendix A).

4.4 Function Analysis

For the design of the test rig a function analysis was conducted to determine which functions the test rig has to offer. The main functions were; damage mechanism within accelerated time and test the mechanism. Among the functions, examples on sub-functions were; simulate forces, offer automatic operation and offer easy use of the rig. The entire function analysis can be found in Appendix B.

5 Concept Generation

The concept generation was made with the requirement specification as the guiding framework. One key requirement was to keep the rig compact, to allow for transportation of the rig in the laboratory. Therefore, the early stages of ideation focused on the overall structure of the test rig and on addressing the questions: "How will the adaptive mechanism be mounted within the rig?" and "How can the time to failure of the mechanism be accelerated?". At this stage, hydraulic forces were intentionally disregarded in order to prioritise the simulation of the mechanical forces. Moreover, the impeller could still be evaluated in the CTR to assess damage under hydraulic forces.

A morphological matrix was used to create the overall structure of the rig by generating alternative solutions for each function. Crazy 8 was combined with brain writing 635, the findings were how the core structure of the rig would be, specifically targeting the placement of the drive shaft, motor and adaptive mechanism. A combination of foot-storming and indirect benchmarking was done with the goal of gaining inspiration from unrelated industries. During the exercise, products with desired features for the test rig used in everyday life were examined to identify how they were built. One product examined was the lazy Susan, with a function allowing the lower plate to become fixed, while the upper plate spins. Some of the components identified for use in the test rig included bearings to enable rotational movement and a stationary support structure. Additionally, either a linear motor or a hydraulic cylinder was considered to provide axial movement.

A development of the CTR rig or methods to use it for testing the adaptive mechanism was explored. Some ideas were to instead of continuously adding fabrics to the CTR, other objects such as wash balls or metal could be added in a closed loop to activate the adaptive mechanism repetitively. The idea was dismissed since the impeller would only shred the wash balls to pieces and metal objects would damage the impeller and not target the adaptive mechanism. Another consequence is that the objects would get wedged into the impeller vanes. Moreover, the reason this was not taken further was that the goal with the test rig was to create an accelerated test and it was believed that using different motors to simulate repetitive movement would speed the process up as well as reduce manual labour. Additionally, it would create more repeatability between tests and allow for an easier identification of specific forces impact on the adaptive mechanism.

After the initial structure of the rig had been explored, several concepts for cooling the system were generated. Due to the anticipated high-speed interaction between the stainless-steel components of the adaptive mechanism. It was initially assumed that the repetitive motion and applied forces would generate elevated temperature through friction. Consequently, different cooling approaches were considered, including water, oil and air-cooling systems. Air cooling relies on airflow to dissipate heat, whereas liquid cooling systems, use fluid to transfer heat more effectively (Arctic, 2025; Nedelcu, 2024). Although air cooling systems are generally simpler to install and eliminate the risk of leakage, water cooling offers a higher thermal capacity, enabling more efficient heat absorption and therefore improved cooling performance. Oil cooling was also considered due to the higher boiling point of oil, which makes it suitable for applications involving elevated temperatures. However, oil cooling was regarded as less favourable because of its lower cooling efficiency compared to water, as well as the potential fire risk associated with its uses (Nedelcu, 2024).

The following concepts were the initial concepts of how the overall structure of the test rig would look like. The initial concepts included the motor's position in relation to the impeller, major components required for the test rig, and water-based cooling systems. Other details such as safety systems and sound vibration will be determined later, since it can be adapted to any type of structural test rig.

5.1 The First Concept

The first concept contains a linear-rotary motor capable of both rotational and axial movement; see Figure 8. The motor is positioned beneath a water filled container with the impeller on the motor shaft, allowing the continuous water flow to cool the parts of the adaptive mechanism. Compared to the natural orientation of the impeller in the wastewater pumps the impeller in this concept is upside down. A lid is mounted on top of the container and applies axial pressure to the end of the impeller. To prevent the rotating impeller causing fiction welding or drilling into the lid, the lid contains a thrust ball bearing where the impeller connects. During operation the impeller presses against the axial ball bearing as the shaft within the impeller is moved axially by the motor, compressing the spring continuously. This recreates the impeller movement created as large objects pass through the wastewater pump, by moving the shaft within the impeller, instead of the impeller on the shaft. The spring is compressed and the splines on the sleeve collide with the surrounding components and interact with the insert. The design allows the impeller to be mounted on different lengths of the shaft to allow for different impeller models.

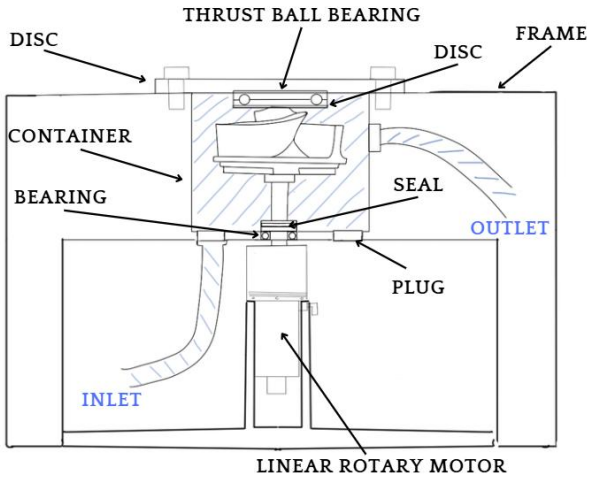


Figure 8 The first concept with arrows showing different components

The test rig is assembled by placing the motor and the shaft underneath the container. The axial ball bearing is screwed into the centre of the lid to align with where the impeller will connect. The inlet allows for water to flow into the container and out through the outlet. When the test is finished, water can be emptied by removing the plug. Advantages of the first concept is; simple assembly process for the impeller and lid. The impeller is continuously cooled due to the water flow. Noise can be contained due to the impeller being completely sealed within the container. However, the motor is positioned directly underneath the water filled container, meaning a leakage could lead to motor failure. Another potential issue is vibrations that would arise, due to the impeller pressing against the bearing attached to the lid. Lastly with one motor recreating both the linear motion and rotational motion it is not possible to adjust the input frequency of the rotational and the linear movement separately.

5.2 The Second Concept

The second concept consists of two motors, one that is connected to a disc to recreate linear movement, and one attached to the impeller to generate rotational motion; see Figure 9. The linear motor can be moved along a groove and locked into place avoiding movement during testing, and also for easy assembly. The impeller is mounted on the shaft of the rotational motor, the linear motor is moved along the grooves to connect with the impeller. In this concept, the impeller has a horizontal orientation, however that should not affect anything since Xylem today has wastewater pumps that are installed horizontally.

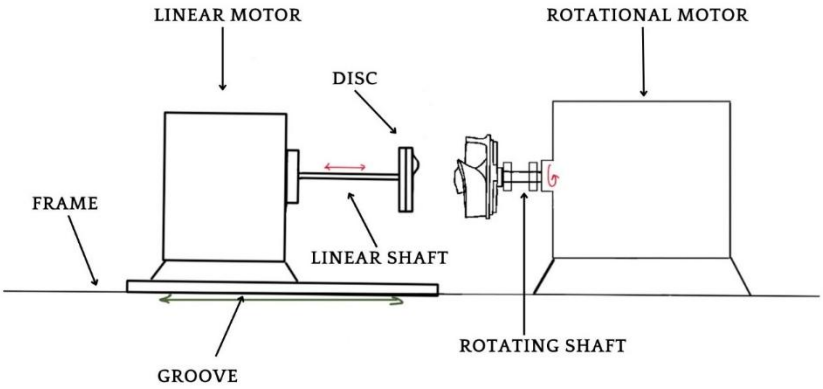


Figure 9 The second concept with arrows showing different components

During operation, the rotational motor rotates the impeller at high speed, while the linear motor generates a pulsating axial motion through the disc. As a result, the impeller repeatedly moves axially from its distal rest position to its proximal position and back again. To avoid any damage created between the linear shaft and impeller, a disc containing thrust ball bearings will be mounted on the linear motor shaft. This disc will be in constant contact with the impeller during the test. By designing the contact surface of the disc acting on the impeller with varying geometries and tilt angles, the test setup could incorporate controlled impeller misalignment. This would test the motion that happens when an object passes through the impeller. The disc that will be in contact with the impeller can be changed when needed. The second concept does not contain water and is only operating in air.

The advantage with the second concept is the lack of water since it simplifies the use of the test rig. It also allows the test rig to be used in an environment where there is no water source in the laboratory. The disadvantage with the second concept is that rotational motor must be able to withstand the force that the impeller will have to simulate. Furthermore, since the linear motor is installed on grooves for easy assembly, there is a risk of the motor relocating under the given force.

5.3 The Third Concept

The third concept use pressurised water to simulate the forces the adaptive mechanism is exposed to; see Figure 10. The concept is inspired by washer pressures that uses pressurised water to clean surfaces and remove debris. In this concept, the impeller is horizontally oriented.

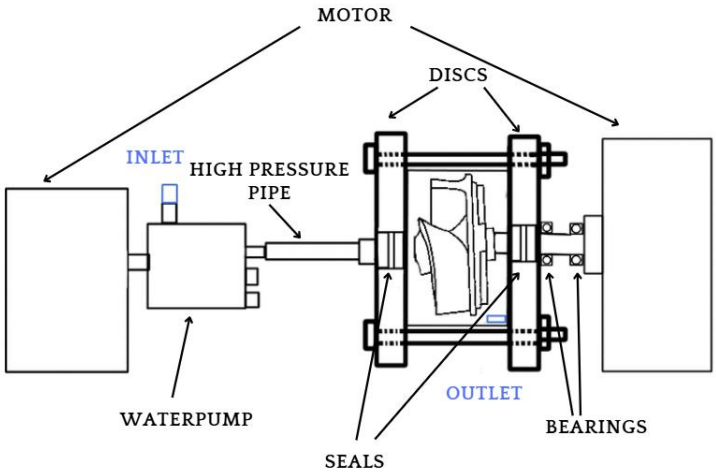


Figure 10 The third concept with arrows showing different components

On the drive shaft of the rotational motion motor a disc would be attached before the impeller is screwed on the shaft. This disc is then connected to a cylindrical container and another disc to enclose the impeller. Both discs contain seals to ensure no water leaks out and the high-pressure pipe would be connected trough one of the discs. A motor runs the water pump to create pressurised water, pushing the impeller into its proximal position. Due to the high-pressure watering being transferred through a pipe it allows for different pressure points on the impeller. By designing interchangeable variants of the left disc, the positions of the inlet hole for the high-pressure pipe can be altered. This allows the impact point on the impeller to be shifted, thereby introducing controlled pressure variations and potential misalignments during operation. The water flows out of an outlet in the first disc close to the impeller shaft. The hole in the impeller eye would be blocked during the test to create a larger surface for the force to work upon. Since the impeller is going to move from the distal rest position to the proximal position, the water pressure will be automatically adjusted to allow the impeller to return to its distal rest position. Furthermore, the water flow replicates the vibrations to which the impeller is exposed in its actual operating environment. The continuous water flow will also cool the adaptive mechanism during operation.

The advantage with the third concept is that the pressure point from the water pressure can be altered to different positions. It is also possible to change the waterflow by having different diameters on the high-pressure pipe. Moreover, having different containers also ensures that different impeller models can be tested, since they vary in height. It is also possible to decide different frequency for that axial and the rotational movement due to them being controlled with separate motors. However, the concept requires automatic control to turn on and off the water, to allow the impeller to return to its distal rest position.

5.4 Feedback On The Concepts

A feedback session was held with relevant engineers from Xylems R&D team to present and evaluate the feasibility of the concepts. The selection of parameters and how these would be measured for the test was also discussed. It was agreed to define these after the concept had been decided, although some initial suggestions were made. One suggestion was to use a stroboscope to measure the rotational speed of the impeller. To measure the distance the linear motors needed to push the impeller, load cells could be connected to the motor. It was suggested that load cells also could be used to identify the forces working on the impeller for the third concept, since it would be difficult to predict the pressure within the cylindrical container.

Concerns were raised about the third concept, as the complexity of the test rig increases when water pressure is included. The water flow introduces unpredictability and turbulence, which may complicate the interpretation of measured data and the identification of the causes of damage to the mechanism. In addition, this variability makes it difficult to reproduce damage consistently, which limits the comparability with real operating conditions. Further research into the hydraulic forces indicated that replicating the real-world condition within the proposed concepts would be challenging. Moreover, it was mentioned that the test rig potentially does not require water, since the radial and axial force could be replicated without the hydraulic forces. A previous concern regarding excessive temperature rise in the adaptive mechanism was dismissed by one of the interviewees. Although the pump operates at rotational speeds between 1000-4000 RPM, which may result in heat generation in the motor. A senior designer engineer suggested using the drive unit for the impeller model being tested, as these units have integrated cooling systems thereby eliminating the need for any additional motor cooling system. Another advantage of choosing an existing motor from Xylems wastewater pumps was to avoid the need to source and dimension a separate motor capable of achieving the required rotational speeds, as well as designing a compatible drive shaft.

Additionally, a senior hydraulic engineer was consulted on the different concepts. One thought was that it was important to simulate the force from the objects that creates misalignments on the adaptive mechanism. For both the first and second concept, it was advantageous to have an uneven surface on the disc that would be in contact with the impeller. The disc could be redesigned to test different angles of misalignment for different tests. Another interviewee mentioned that to create a large misalignment, the plate in contact with the impeller should be pressed as close to the outer circumference of the impeller as possible. Furthermore, since the impeller does not have a flat surface on the bottom this presented an issue. Because all the components within the same pump series have the same adaptive mechanism, the test can be conducted on the one impeller with the largest outer diameter (LT) in order to create a critical misalignment.

Furthermore, to prevent frictional contact and potential friction welding between the rotating impeller and the stationary linear actuator shaft, bearings were evaluated. Since the impeller would rotate at 3000 RPM, the bearing could address the friction welding. One solution was to mount an axial bearing on the shaft, allowing one side of the bearing to rotate together with the impeller. Different axial bearings were considered to withstand the rotating speed and the impact load generated during collisions between the impeller and the linear actuator shaft. However, several issues were identified in using an axial bearing. The bearing would be directly exposed to the impact loads without extra support. This could result in damage in the bearings, since they are not designed for that. Furthermore, the bearing would need to accelerate rapidly during each impact. This would cause mechanical stress and even heat. Moreover, another challenge with the intended bearing solution was that axial bearings typically require fixation on both sides. This became problematic, as the bearing assembly also needed to translate axially together with the linear actuator shaft during collisions with the impeller. As a result, a conventional fixed-end mounting configuration was difficult to implement while still allowing the required linear motion. Although, inspiration for a solution that would replicate the axial bearing's function was inspired from the feasibility test in the lab.

In addition, another comment about the disc was about whether the disc should be in continuous contact with the impeller or collide with it. The initial idea was that the disc would follow the impeller, however this does not fully replicate the punching force created when objects collide with the impeller in real operation. It was suggested to replace the disc after every use, despite this, the material for the disc would be crucial to avoid it being damaged before the test is completed. It was also noted that the damage that the cover is exposed to would affect the damage on the impeller, since the disc might deform during the test. The interviewee recommended materials with favourable sliding properties such as brass, rubber and polyurethane. Although due to the impeller being in a dry environment with no liquid, rubber or similar, it would run the risk of overheating with constant contact.

Together with this, two suggestions were made for the linear motor, a pneumatic cylinder or a step motor. The advantages with the step motor were its high torque and quick acceleration. The first concept was believed to be feasible, if a favourable motor for both the rotation and linear movement could be found. Regarding the second concept, one concern was the weight of the motor creating the axial movement. For it to not move, it would need to be heavy. It was also questioned whether the impeller must rotate at the same speed as in real conditions or at all and should instead only be exposed to axial collision instead of the combination of them both. An overall comment for all the concepts was that since the impeller will rotate and the aim is to damage the mechanism; the rig must have a strong safety cover. Avoiding broken parts flying away, or people getting hurt on the impeller in motion.

5.5 Evaluation Of The Concepts

The different concepts were compared and weighted towards the requirement specification, found in Appendix A. The criteria weighing method was used, which can be seen in Appendix C. The method resulted in ranking the concepts highest to lowest, second concept, first concept and third concept. The value between the third concept and the other concepts was large and therefore it was not taken forward. Moreover, the use of high pressurised water would make the test rig highly complex since it would require sealings, additional components for pressurised water and a water source. Additionally, the engineers stated that the damages created by pressurised water would not be realistically occur under actual operating conditions. There was no marginal difference between the first and second concept, meaning the final choice was based on other factors.

After further evaluation of the first concept, it was evident that the linear rotary motor would not be able to simulate the high forces needed for the axial movement. An alternative motor mechanism using a ball screw was then evaluated. However, it was disregarded due to the complexity of the required drive shaft design. The shaft would need to integrate multiple functional interfaces, making it difficult to manufacture and impractical to produce separately for each impeller model. Therefore, the first concept was not taken forward. The second concept was selected for further development, as it was considered more suitable in comparison to the alternative. Since the second concept does not incorporate water, and feedback revealed that cooling of the adaptive mechanism was unnecessary, the study focused on implementing misalignment and linear motion in the test setup and not on any hydraulic forces.

6 Development

The following chapter describes the development of the chosen concept and the simplified prototype testing.

6.1 Concept Description

The chosen concept was named repetitive movement test rig (RMTR), referencing the aim to create repetitive movement in the adaptive mechanism. In real operation the impeller rotates creating a pressure that draws water through, as fabrics and other objects passes, they create a large force that pushes the impeller upwards. To simulate the physical movement of both rotation and linear collision there will be two different motors in the test rig; see Figure 11. On one side a drive unit from one of Xylems existing water pumps, will be used to rotate the impeller. On the opposite side there will be a linear motor creating the impacts loads. In order for the linear motor and impeller to interact without friction welding, there will be a cover attached to the impeller and a disc attached to the linear motor shaft. On the disc an axial bearing will be attached. The disc will contain different geometries to recreate misalignment in the mechanism.

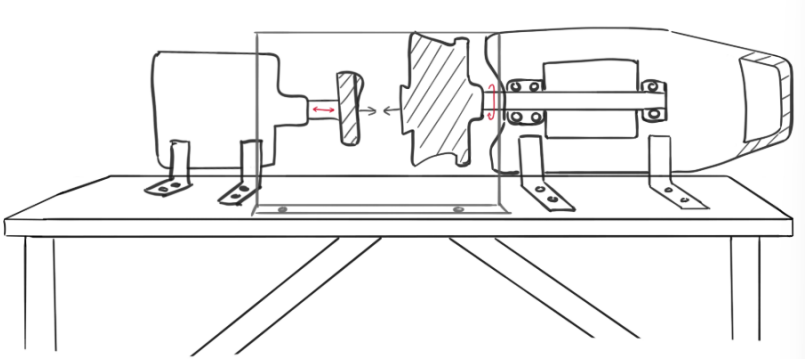


Figure 11 Iterated version of the selected concept

6.2 Cover Of Impeller

The high rotational speed of the impeller introduced significant safety considerations, leading to the development of a cover attached to the impeller. In addition to the safety considerations, the geometry of the impeller is not flat at the bottom, complicating the application of the misalignment motion. The cover was consequently used as an interface to enable controlled force application and introduce misalignments. Moreover, interchangeable plates were decided to be integrated into the cover to create varying levels of misalignments, allowing the same cover to be reused for different tests. However, one concern was if the cover could dampen the applied forces or fail under repeated impacts during testing.

The cover was initially created using the casting mould of the impeller, and therefore no separate attachment mechanism was integrated into the design, as seen in Figure 12. However, after testing the cover, different attachment solutions were explored to improve stability during operation. Due to the limited space above the impeller, the attachment needed to be compact and not complicate the assembly of the impeller onto the drive shaft. The concepts were compared in Creo Ansys to determine the most suitable solution based on the forces acting on the cover.

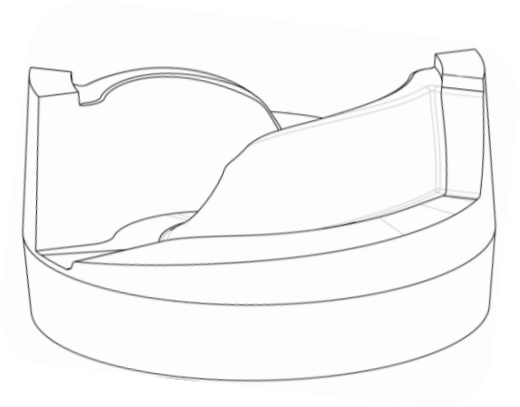


Figure 12 Model of cover for impeller

Material selection was critical to ensure that the cover could withstand repeated compressive loads and high rotational speeds without excessive deformation. Initially, polylactic acid (PLA) prototypes were found lacking sufficient strength and, a mix of plastic and metal design was also rejected due to limited robustness and increased manufacturing complexity. Several materials were compared in Granta EduPack based on hardness, density, and yield strength. The materials included stainless steel, aluminium, bronze, brass, polytetrafluoroethylene (PTFE) and polyoxymethylene (POM). Stainless steel was excluded due to the risk of damaging the stainless steel axial bearing on the disc. In addition, weight was a critical constraint, since the cover should be manually assembled onto the impeller.

Finally, concepts for introducing controlled misalignment through the cover geometry was explored. Surface features such as bumps and raised geometries on the bottom surface of the cover were considered to force the impeller into angular displacement as the bearing connected with the surface.

6.3 Test In The Laboratory

Evaluation of the feasibility of the impeller cover happened by mounting an impeller with a cover onto the shaft of a 6020-drive unit and tested at 50 RPM. However, the cover was tested without an additional attachment securing it to the impeller. During the first minute of operation the cover remained in place, although noticeable wobbling was observed. After the test was stopped and restarted, the cover detached and fell off immediately, caused by the initial rotation back and forth as the pump started.

Another test was made to determine if the test was feasible, a simplified version was developed in the laboratory. A crankshaft was attached to the impeller shaft to be able to rotate the impeller by hand. On the opposite side a lathe centre was mounted with a screw at the end, allowing the lathe centre to be moved linearly. The impeller moved back to the distal rest position when the lathe centre was removed because of the spring in its mechanism. A shaft with a bearing was placed on the side of the lath centre to recreate the misalignment by moving the bearing shaft linearly and letting the bearing push on the plate. The cover was complemented with metal plates to secure the impeller inside the cover and reassure that the cover stays on during the test; see Figure 13.

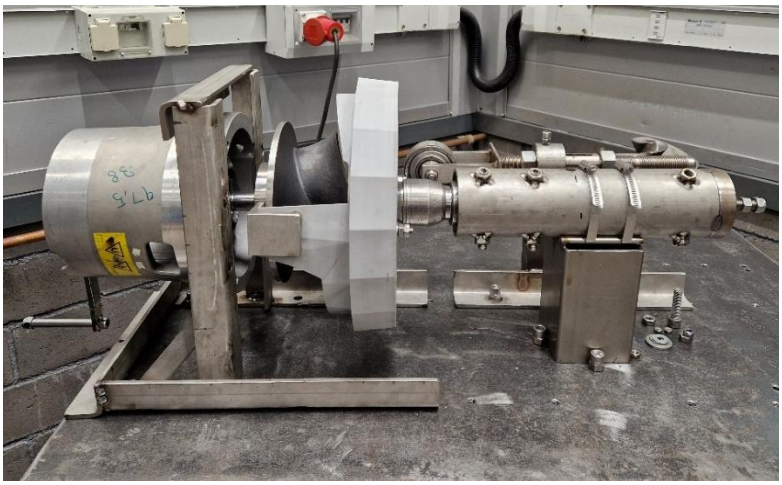


Figure 13 Simplified version of the RMTR, front view

The first test conducted was to determine if misalignment could be tested simultaneously as the linear movement. The bearing shaft was placed to push on the impeller synchronously as the lathe centre moved the impeller linearly. It was quickly realised that the bearing did not create a significant misalignment at the same time as the linear movement occurred. The bearing only facilitated with moving the impeller axially. Because of these findings the test was divided into two separates tests, one for linear movement and one for misalignment. To create the misalignment the impeller needed to be pushed as far from the centre as possible. Ensuring that the impeller travels along the full length of the splines requires the force to be applied at the centre of the impeller. This was positively received by the engineers interviewed and the users of the test rig, as it would simplify identifying what damage the different movements cause.

The linear movement test was conducted both with and without the cover by letting the lathe centre move the impeller axially; see Figure 14. There were no issues with the linear test both with and without the cover and for different rotational speeds. It was remarked that without the cover it would not be possible to conduct the linear test on the HT impeller as the sleeve extends outside the impeller when in the proximal position. It was realised the lathe centre was a good component to allow for one rotating part and a stationary shaft. This removed the issue with choosing an axial bearing as the lathe centre is already dimensioned for similar movement. During testing, issues arose when contact between the rotating impeller and the lathe centre was removed and then re-established, to simulate collisions. While, the lathe centre rotated simultaneously with the impeller during continuous contact, it was unable to accelerate sufficiently to match the speed upon reconnection within the required time frame. This led to the decision to keep the impeller continuously pressed against the lathe centre throughout the entire test.

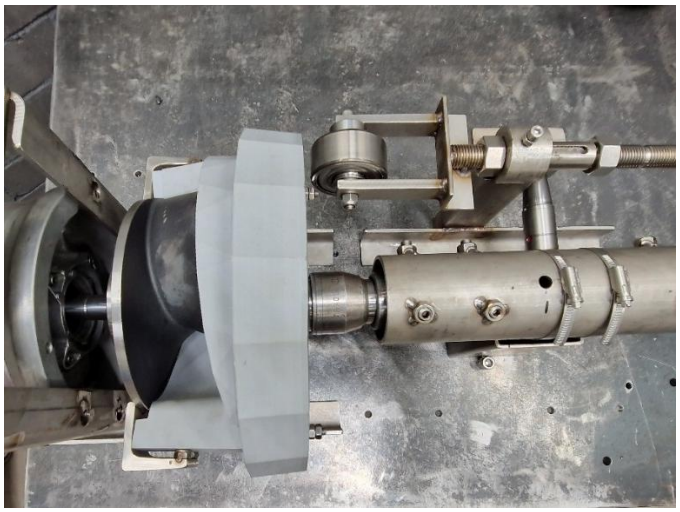


Figure 14 Simplified version of RMTR top view

The misalignment was tested by compressing the spring in the impeller by pushing the lathe centre into the proximal position. The bearing was then placed to connect with the impeller, and the lathe centre was released, to create a bending force. The impeller was rotated and the finding was that the impeller did not move in any angle.

The misalignment was also tested by applying a hitting force on different positions to the bottom of the impeller cover. The force was first applied at the largest radius of the impeller, then the outer vane, and then close to the centre. The largest misalignment was caused at the largest radius 101mm. The same test was conducted when the spring was compressed to different lengths, to see if the misalignment was larger when the spring was fully, partially or not at all compressed. The finding was that the misalignment was similar between all positions, but slightly larger when the spring was not compressed at all.

6.4 Mock Impeller

It was decided to discontinue the cover and instead redesign the exterior of the impeller. The decision was based on several insights gained during the physical testing in the laboratory. During laboratory testing, one of the challenges was securing the cover on the impeller. The initial idea assumed an inherent geometric fit between the impeller and cover. However, during laboratory testing when the impeller was rotated, it was discovered that there was a gap between the impeller and cover. As a result, during the misalignment test the cover became misaligned and the impeller remained unaffected as it was not fully connected. Additionally, the cover added unnecessary weight to the motor shaft.

Moreover, a meeting with the supervisor revealed that a simpler version of the impeller could be cast at Xylem's foundry in Sweden. This simplified the material and manufacturing choice compared to using a cover, since the test version of the impeller could be made with the same material as regular impellers, in grey iron. The interior of the mock impeller was made identical to the actual impellers, to ensure that the adaptive mechanism can be assembled. In order to apply different forces to the mock impeller, the exterior bottom surface required to be flat. This version of the impeller would be designed for the requirements of the RMTR and not be included in other test rigs such as the CTR.

To transfer the force from the linear motor through the lathe centre to the mock impeller, the contact surface of the mock impeller was designed with an angled geometry. The lathe centre has a 60° angle, and to prevent self-locking, the angled geometry was decided with a contact angle of 61°.

Different alternatives of mock impellers were explored to determine a suitable version that would meet the requirements. The mock impeller needed to be lightweight, sufficiently durable to withstand the applied forces, easy to assemble onto the drive shaft, and designed without sharp surfaces that could pose a risk of injury. One of the main concerns with using the actual impeller in the RMTR was the safety risk posed by sharp vanes rotating at 3000 RPM. Therefore, safety was a key consideration in the design of the mock impeller. Multiple variants of the mock impeller were created; see Figure 15.

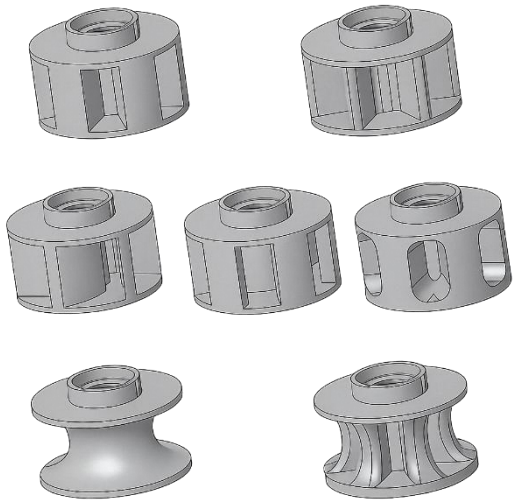


Figure 15 Seven different versions of the mock impeller

The various version was analysed in Creo Ansys to determine which geometry would be least affected by the applied forces. The simulation evaluated the misalignment of the components and von Mises stress. However, it was not a marginal difference between the different version; see Figure 16. Therefore, the criteria weighting method was applied to compare the designs against the key requirements of safety and weight, see the entire result in Appendix D.

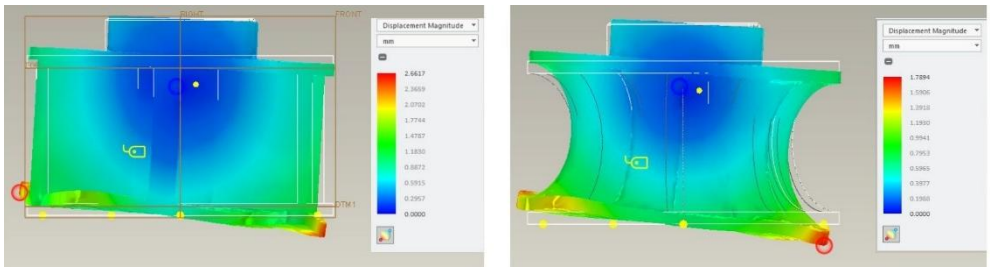


Figure 16 Displacement magnitude from Creo Ansys, left shows impeller with straight vanes, right shows impeller with biconcave vanes

With the method, two options were top performing; one with thin straight vanes and one with biconcave shaped vanes. The final selection was based on the better Creo Ansys results, together with input from the industry supervisor, see Figure 17. The material for the mock impeller was chosen as grey iron, which is the same as the existing impeller. Moreover, to preserve the lathe centre from damage and be able to use it for multiple tests a disposable adapter in a softer material was developed. This was done in order to allow only a smaller component to be replaced, thereby avoiding the need to replace the lathe centre. However, after consideration, this solution was discarded as a mock impeller would be manufactured for each test to accommodate design changes of the adaptive mechanism. Furthermore, since one of the components is

permanently integrated into the mock impeller during manufacturing, the mock impeller cannot be reused between different test iterations. Additionally, the reason for excluding the adapter was that it was intended to be made from bronze, raising concerns that it could deform during testing and lead to inconsistent results over time.

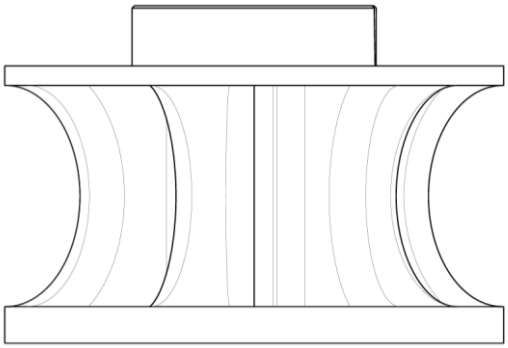


Figure 17 Final design of mock impeller

6.5 Workbench And Frame The Test Rig

Feedback on construction of the RMTR rig was received from a senior design engineer. One concern was the weight of the workbench and frame, since most of the components were intended to be made out of steel. However, it was quickly realised that the complete test rig would be extremely heavy and difficult to move around in the laboratory. Therefore, it was recommended to re-design the workbench and frame, by having hollow legs, a thinner tabletop and lighter supports for the drive unit; see Figure 18 for comparison between first workbench and final version. It was also recommended to change from steel to aluminium. The design engineer further explained that reducing material thickness carries the risk of structural instability, while increased thickness results in unnecessary weight. It was also discussed if the linear guide rails would manage being exposed to the force. However, since there are two linear guide rails, the structure becomes more secure.

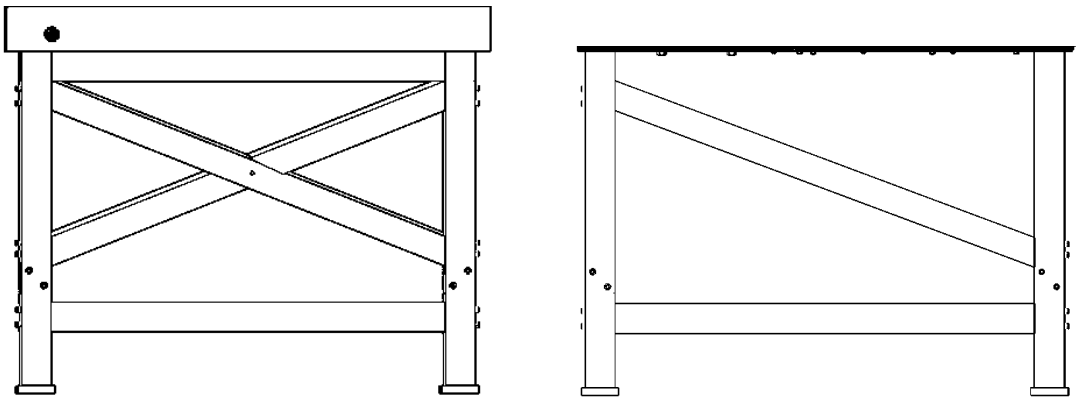


Figure 18 First version of workbench on the left, final version of the workbench on the right

The idea was that the safety barrier would be made from aluminium, and have a window made of plastic. However, the metal of the safety barrier would amplify the sound the test would create. It was also noted that one of the existing test rigs in the lab used plastic as a safety barrier. The safety barrier in that test is used to protect the user from injury as well as from any parts that comes loose. It was thus decided to use

plastic for the safety barrier. The benefit with plastic is that the material can be attached together by using adhesive, which allows for necessary design choices. It was explored if the safety barrier would be manually lifted from the RMTR and then reattached between each test. However, it would be unwieldy because of the large dimensions and therefore it was disregarded. It was also discussed if there would be a gas spring, to aid for opening the safety barrier. Despite this, the weight of the barrier was not substantial and therefore a string fastened to stop the safety barrier in the right position would be enough, see Figure 19.

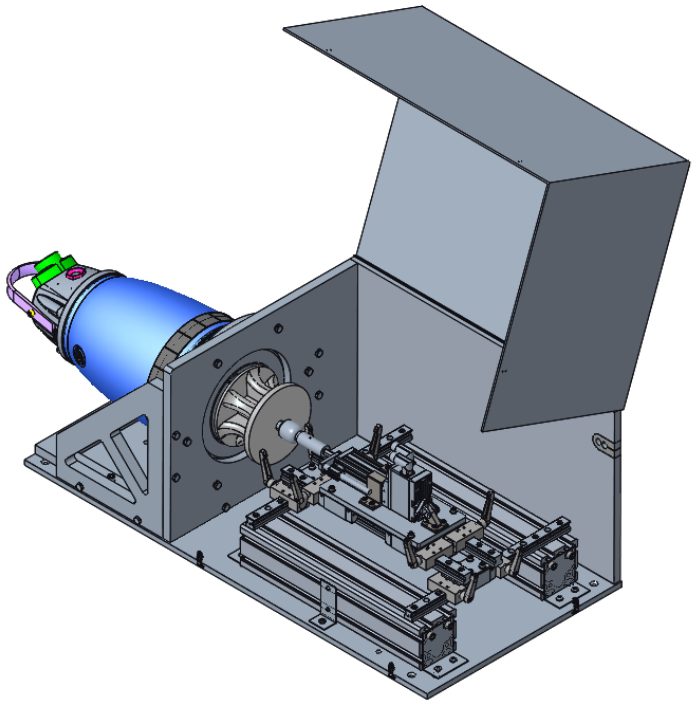


Figure 19 All components of RMTR with open safety barrier

The linear actuator has two cables which will have to be guided away from the rotating impeller. Thus, it was decided to have a hole in the back wall to connect the actuator to necessary equipment. To reduce the sound a foam, insert into the cable holes was recommended by a senior design engineer; see Figure 20. This would allow the cables to naturally expand the foam into the hole in the back wall to help with dampening the sound.

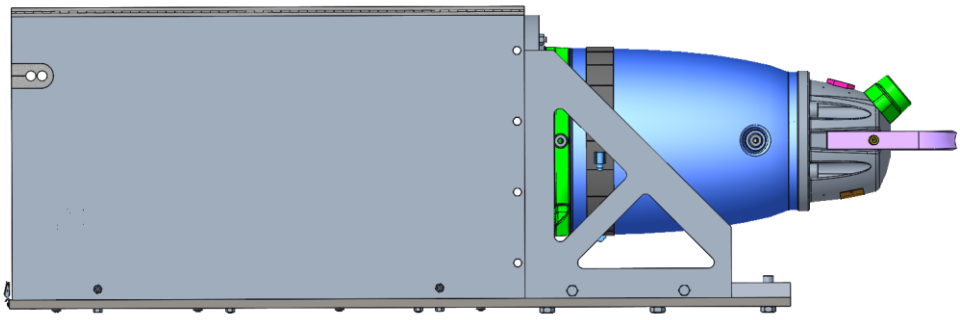


Figure 20 Back view of the test rig with foam insert

7 Final Design

The following chapter explains the exact components and parts for the RMTR, as well as the final version of the concept.

7.1 Choice Of Motor For Rotational Movement

Based on the feedback and further iteration of the concept, the use of Xylem's Concertor 6020 drive unit was evaluated through an interview with a development engineer in the motor team. The test would require the impeller to rotate at 3000 RPM, and this was confirmed not to be an issue. The engineer also noted the motor would unlikely be damaged under the applied linear loads, as these are low in comparison to the forces in real operating conditions. The Concertor includes a smart anti-clogging function. However, a potential risk is that this function could be activated by repeated collision events, causing the impeller to reverse direction and thereby affecting the test result.

Another concern was related to the flexibility of the RMTR, since the purpose of the setup is to evaluate future impeller iterations. The Concertor was considered more advanced and costly than necessary, as the test does not focus on the motor. The motor engineer therefore suggested using the 3153-drive unit instead, since it also included an integrated cooling system. This simplified the setup, as only the shaft needed to be modified between different impeller designs, making testing more efficient.

7.2 Choice Of Machinery For Linear Movement

The chosen actuator that will provide the linear movement is the Tritex TTX060-080-04-D-A-N-D-SIO-5-A from Curtiss Wright. It is a servo-driven linear actuator that includes an inverted roller screw transmission (Curtiss Wright, n.d.). The servomotor generates rotational torque, that is transmitted through the roller screw mechanism. Due to the threaded contact between the screw and rollers, the rotational motion is transformed into linear motion of the output shaft. The linear actuator has an anti-rotating rod that ensures that the rod in the actuator does not rotate, due to the roller screw. The linear actuator also includes a break to be able to stop the motion instantly in case of emergency. The actuator meets the requirement for operating the test rig. Specifically, it provides the required minimum stroke length, delivers a continuous force, and achieves a full back-and-forth motion within a short timeframe. The duty cycle was required to be continuous, meaning the actuator would operate without scheduled pauses or rest periods, since the intention was for the actuator to run continuously during operation. Additionally, it was important that the actuator's stroke length be adjustable to meet varying requirements for different test. For the chosen actuator, it is possible to move the linear movement with 1 mm length, which is considerably precise.

The reason for choosing a linear actuator is that it has similar properties as hydraulic cylinders, without needing external components. The pneumatic cylinder was otherwise an option; however, the extensive components were unwanted. The linear actuator allows for different input, such as stroke length, speed and force. The actuator can operate in two modes: force-controlled or displacement controlled. In displacement mode, the user specifies a desired stroke and speed, for example instructing the actuator to move 20 mm in 0.5 seconds, without needing to define the force. Alternatively, in force-controlled mode, the user sets a target force, e.g., 700N, and the actuator moves until this force level is reached. This enables changing the amount of force applied or number of impulses, to test different cases and se when the mechanism sustains damage.

7.3 Linear Test Structure

After the feasibility test, it was decided to use a lathe centre instead of axial bearings between the mock impeller and linear actuator shaft. The lathe centre is designed to be used for objects rotating at a high speed and is capable of handling the pressure required to securely hold the object in place, while it is subjected to radial forces. The criteria for the lathe centre in this project were similar, as it needed to withstand the rotational speeds of up to 3000 RPM and the axial forces generated during the movement of the mock impeller. Another consideration was the mounting of the lathe centre onto the linear actuator shaft. Lathe centres are typically manufactured with a tapered end (morse taper) for assembly into a lathe spindle. For this application, a smaller morse taper was selected to ensure compatibility with the linear actuator, MT3 was chosen as the appropriate size. Moreover, the final decisions were between a shock-absorbing model and a lightweight option. Since impact collisions were removed from gathered information, the lightweight option was ultimately chosen.

To assemble the lathe centre with the selected linear actuator, several interfacing components were developed, see the configuration in Figure 21. One of the components is a rod extender and is an industry standard. The reason for using a rod extender is that the length of the shaft is rather short. The rod extender consists of a cylindrical threaded component with an external hexagonal profile with an internally threaded bore. On the linear actuator shaft Loctite will be used before screwing on the rod extender. Loctite works as an adhesive that hardens when there is no air, which happens when other components are assembled onto it. The rod extender will then function as a coupling between the actuator and the component holding the lathe centre, while also maintaining the stiffness of the shaft.

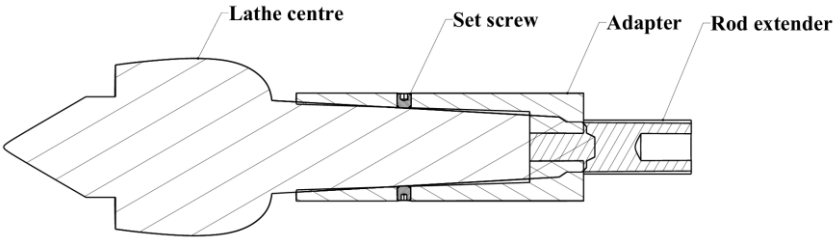


Figure 21 Cross section of the lathe centre, adapter, rod extender

Moreover, to install the lathe centre, a morse taper socket was required. However, mounting the morse taper socket directly onto the rod extender was not feasible. The reason being that the socket relies solely on set screws for fastening, which would allow for misalignment during testing. To address this, a custom adapter was developed to connect the lathe centre to the linear actuator, as part of the overall assembly system. The adapter is a cylindrical component with an internal thread, allowing it to be securely attached to the rod extender. Internally, it features a tapered geometry matching the morse taper of the lathe centre. Additionally, two threaded holes for set screws are included to ensure that the lathe centre remains securely fixed within the adapter during operation. These set screws are particularly important due to the horizontal pulsing motion, especially when the lathe centre is pulled back by the actuator. The adapter is designed for an MT3 lathe centre, accounting for the associated dimensional requirements. Equipment changes are simplified by unscrewing the adapter with the lathe centre attached and replacing it with an alternative shaft, such as one equipped with a bearing for the misalignment test.

7.4 Misalignment Test Structure

For the misalignment test, other components were required to be mounted onto the linear actuator shaft. In this setup, a bearing rolls along the underside of the mock impeller while the linear actuator applies an axial force, thereby inducing misalignment. The bearing is mounted on a shaft, which is connected to a holder. The holder is welded to an M8 threaded shaft, allowing it to be screwed into the rod extender for secure attachment; see Figure 22.

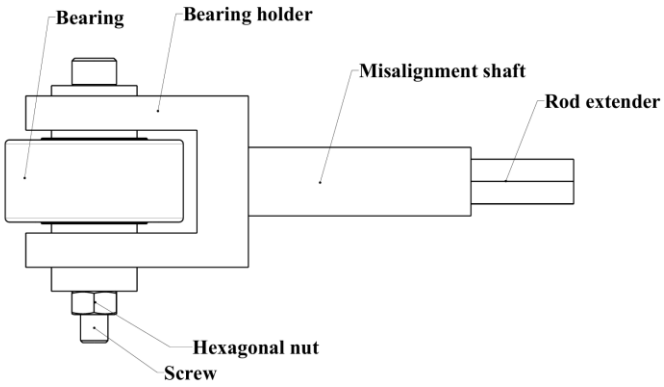


Figure 22 Misalignment component, bearing-holder

The forces the bearing is being exposed to is large, and SKF was contacted to help choosing a suitable bearing. It was recommended to choose a support roller, yoke-type track roller. The PWTR 25.2RS was selected since it is dimensioned for cam mechanism and rolling against planar surfaces (SKF. (n.d.). It tolerates high radial load and can accommodate axials loads due to tilting. The tilting can occur during the testing, as the rotating flat surface can generate contact forces that cause the bearing to shift slightly toward the centre. The bearing is also dimensioned for high rotational speed up to 3200 RPM (SKF, n.d.). To choose the bearing with a suitable diameter, different variants was explored and compared in relation to the speed that would occur where the bearing touches the flat impeller surface. The speed of the contact area would be high, since the bearing is close to the outer circumference of the mock impeller. It was decided that the outer diameter of 52 mm and the outer ring width 24 mm would meet the requirement. Moreover, the material of the component had to be considered, the surface of the mock impeller will be grey iron, while the bearing is made from steel. Which is favourable since the hardness for steel is significant compared to the hardness of grey iron, this leads to the mock impeller experiencing wear before the bearing. This is ideal, as the bearing is intended for repeated use across multiple tests runs.

A lubricating oil film is required between the bearing and the flat surface to ensure proper operation and extend the bearing's lifetime. This also helps maintain rolling contact and prevents the bearing from sticking to the surface of the mock impeller. As the impeller is cast, some surface unevenness is expected, which may induce vibrations or lead to increased variability in the results. To minimise these effects and improve contact conditions, the surface of the mock impeller should be polished.

7.5 Choice Of Material

The software Granta EduPack was used to select materials for the various components of the test rig. Figure 23 presents a comparison of different metals based on density and Young's modulus. Since the test rig is required to be stiff, and it is preferable for materials to deform plastically rather than fail in a brittle manner. Therefore, Young's modulus was a key parameter in the material selection process. The tabletop was constructed from steel, as it must provide high stiffness and withstand significant forces. Moreover, the tabletop must withstand the weight of the drive unit and other assembled components. For other structural parts of the test rig, such as the vertical plate, triangular supports and walls, aluminium was chosen due to its favourable strength-to-weight ratio. For the safety barrier, polycarbonate plastic was chosen. This material offers high impact resistance and is self-extinguishing, enhancing safety during operation. An additional advantage of polycarbonate is its transparency, allowing visual monitoring of the test process and helps to confirm if the test is simulating the intended movements.

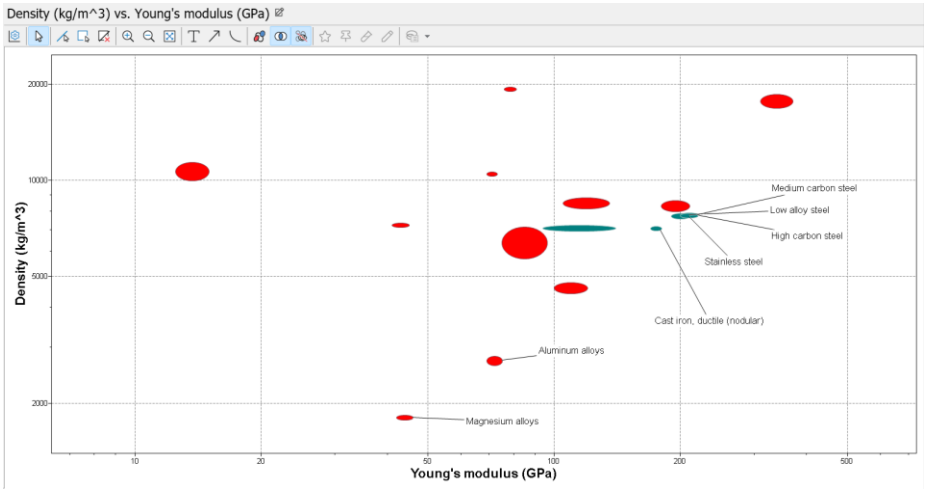


Figure 23 Ashby plot in Granta EduPack

7.6 List Of Components

The components needed to construct the test rig can be seen in Figure 24, excluding drive unit, bearing holder, sensors and safety barrier.

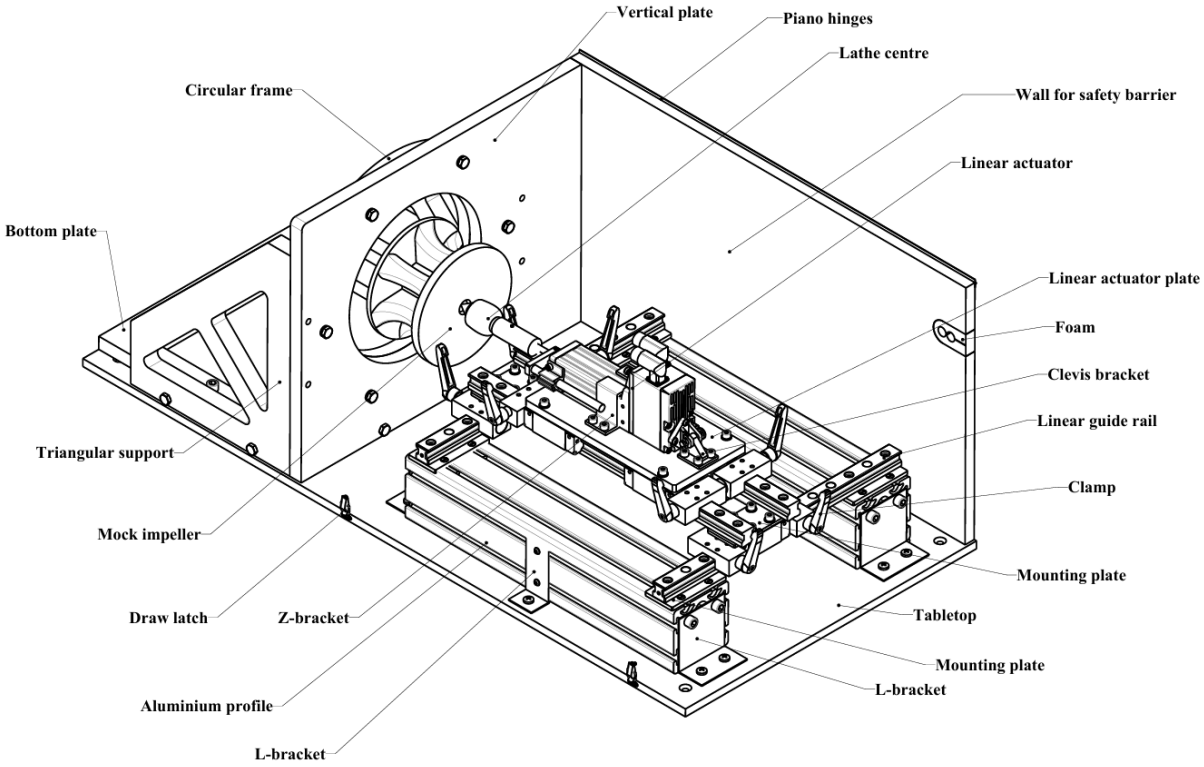


Figure 24 Overall components in the RMTR, excluding drive unit, sensors and safety barrier

The Materials and weight of relevant components are shown in Table 1.

Table 1: Components weight and material of the RMTR

Component	Weight (kg)	Material
Linear actuator TXT	5	-
Drive unit 3153	182 kg	-
Lathe centre	0.95	Stainless steel
Rod extender	0.035	Stainless steel
Lathe centre adapter	0.24	Stainless steel
Misalignment shaft	0.96	Stainless steel
Bearing holder	0.28	Stainless steel
Bearing	0.281	Stainless steel
Z – shaped bracket	0.19	Stainless steel
L – shaped bracket (small)	0.106	Stainless steel
L – shaped bracket (large)	0.229	Stainless steel
Draw latch	9.62×10^{-3}	Stainless steel
Clevis bracket	0.127	Stainless steel
Linear rail (parallel)	5.7	Carbon steel
Linear rail (perpendicular)	4.2	Carbon steel
Rail clamps	0.834	Steel
Table gusset	0.129	Steel
Tabletop	100.2	Steel
Triangle frame	5.6	Aluminium
Circular frame for drive unit	5.3	Aluminium
Aluminium Profiles	9.5	Aluminium
Bottom plate	6.4	Aluminium
Frame drive unit (vertical plate)	15.9	Aluminium
Linear actuator plate	1.9	Aluminium
Mounting plate – rail and aluminium profile	0.124	Aluminium
Mounting plate – linear actuator plate and rail	0.104	Aluminium
Wall (for safety barrier)	22.9	Aluminium
Protective barrier	7.8	Polycarbonate
Mock impeller	12.3	Grey iron

7.7 RMTR Assembly And Structure

The testing procedure will be divided into two separate tests, both conducted on the same table. Xylem's drive unit 3153 will be positioned on one side of the table; see Figure 25. The vertical plate is attached to the bottom plate that in turn is screwed to the tabletop of the workbench. The tabletop features lifting holes at each corner, to facilitate relocation of the test rig with the laboratory's overhead crane. The triangular supports are screwed to each side of the bottom plate and vertical plate. The drive unit is mounted on a circular frame, which is in turn secured to the vertical plate. Within the drive unit, a custom-made drive shaft will be installed to accommodate the selected impeller model. On the drive shaft the mock impeller will be mounted.

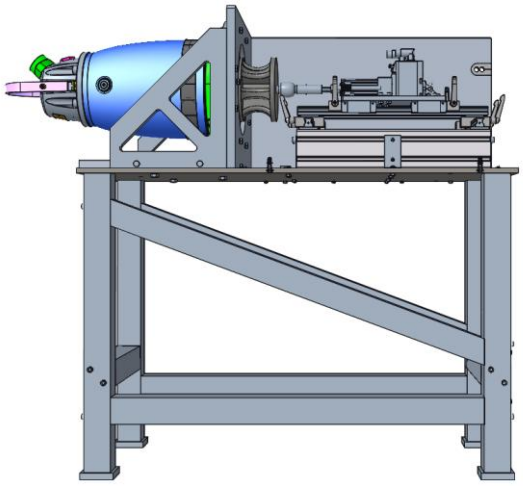


Figure 25 Assembled RMTR without protective barrier

On the opposite side of the setup, a linear actuator is positioned on a system of linear guide rails. This allows motion both parallel and perpendicular to the long side of the table Figure 26. Two aluminium profiles are mounted parallel on the tabletop using L-shaped brackets. The profiles are equipped with T-slot channels, enabling the brackets to be secured with M6 screws and T-nuts. On the shorter side of the profiles, the brackets are attached using pre-drilled and tapped holes, allowing fastening with M12 screws. Two intermediate aluminium plates are mounted on top of the profiles to support the perpendicular oriented linear guide rails. The rails are attached to these plates using threaded holes reinforced with 5 mm heli-coil inserts. These inserts increase the strength and durability of the threaded connections which is necessary due to the limited thickness of the aluminium plates. Carriage mounted on the perpendicular guide rails support an additional set of attachment plates. On these plates a second set of linear guide rails are mounted parallelly. The linear actuator is assembled and then mounted onto the plate by securing a clevis bracket behind the actuator. The Z shaped brackets are screwed into the plate, before the anti-rotate rod is assembled onto the linear actuator.

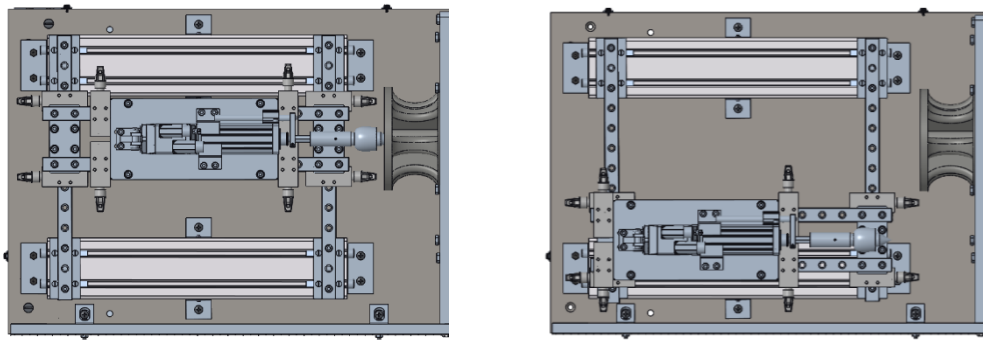


Figure 26 Linear actuator during operation (left), linear actuator during assembly (right) top view

To ensure the linear actuator remains fixed during operation, linear clamps will be installed on both sides of the motor. For the linear test, a lathe centre will be used to engage with the hole of the mock impeller, applying repetitive axial motion via the actuator. In the misalignment test, a bearing will be used to apply

off-centre load to the mock impeller, causing it to move in an inclined manner. The bearing will be in contact with the surface at a distance of 101 mm from the centre.

To enhance safety and reduce noise during operation, a safety barrier will be installed, as shown in Figure 27. In order to further dampen sound from the test, the safety barrier will include D-shaped gaskets along the edges. Piano hinges are used to support the barrier, ensuring even weight distribution and preventing deformation over time. A limiting cord is attached between the barrier and nearest wall to act as a stop, preventing the barrier from swinging too far and striking the back wall when opened. The barrier is secured in place using draw latches, ensuring it remains close during operation.

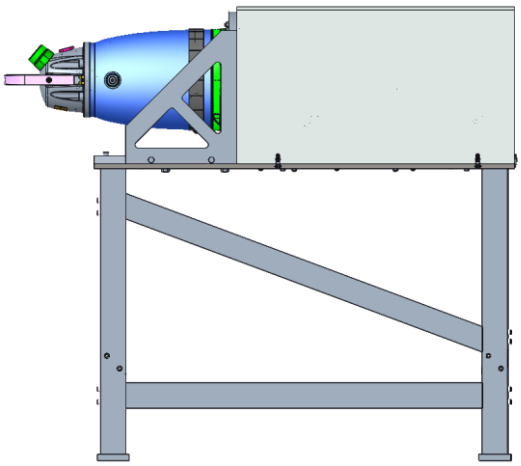


Figure 27 RMTR with safety barrier front view

The parameters that will be measured during the test is drive units power consumption, the vibrations on the drive unit shaft, the temperature of the shaft on the linear actuator, and the number of impulses. The number of impulses is measured by the linear actuator.

The test rig is designed to accommodate both Concertor impeller models 6020 and 6030. Since the adaptive mechanism has identical dimensions for the LT, MT and HT variants, the rig is compatible across all configurations. Furthermore, the setup can be adapted for testing other adaptive mechanisms, with the only required modification being a drive shaft designed specifically for the selected impeller model.

8 Parameters The RMTR Will Measure

For the parameters that would be measured in the test rig, feedback was obtained from a senior development engineer specialising in vibrations. The engineer explained the common measuring equipment available in the laboratory, including its functionality and installation requirements. One key parameter is time, which is defined at the start of the test to set its duration and continuously recorded during operation to indicate when any failure or issue occurs. Moreover, another important parameter is the number of impulses, defined as the number of times the linear actuator shaft compresses the spring within the impeller. This value is measured directly by the linear actuator and provides an indication of how long the mechanism remains undamaged. The number of impulses applied during testing will exceed those experienced under real operating conditions in order to accelerate wear and failure. While this means the results cannot be directly translated to real-life impulse limits, they can be used as a comparative measure between different impellers tested under the same conditions.

In order to identify potential issues within the test rig, particularly at the interface between the lathe centre and the mock impeller, temperature will be monitored with a thermocouple. An increase in temperature, may indicate excessive friction between these components. It was recommended that the thermocouple be placed near the shaft creating the linear motion of the actuator. Since temperature may rise if any issues occur between the linear actuator's attachments and the mock impeller. By alerting to any issues between these components, they can be replaced or reattached before any damage arises in the linear actuator. Measuring the temperature of the surrounding air was not considered effective in this context, as it does not accurately reflect localized heating within components due to its low thermal conductivity and delayed response.

Additionally, the current consumption of the 3153-drive unit will be measured as another indicator of system performance. An increase in current may suggest that components are experiencing resistance or becoming jammed, either at the contact point between the lathe centre and the mock impeller, or within the adaptive mechanism itself. The current is measured internally by the 3153-drive system, which uses integrated current sensors. These sensors continuously monitor the electrical load of the motor during operation.

In addition, vibration measurements will be included to identify potential issues within the adaptive mechanism. Irregular behaviours of components are expected to result in increased vibrations in the impeller shaft. For vibration measurements, an accelerometer was identified as the appropriate sensor. One consideration discussed was the large volume of data generated when continuously logging vibration signal during operation, which can result in substantial data storage requirements. Moreover, it was noted that significant spikes in vibration signals often indicate that damage has already occurred. However, since the test rig operates in a controlled environment by excluding water, vibration monitoring can still provide useful information. For instance, vibration signals can provide useful indications of components deformation, both within the test rig and adaptive mechanism. Together with this, the placement of the accelerometer is critical for accurate measurements. It should be installed near the area and mounted on a component that consistently transfer loads. For example, attaching the sensor to a hollow structure may result in inaccurate data. The accelerometer measuring the vibrations will be installed close to the bearing in the drive unit 3153.

Furthermore, another method considered for fault detection within the mechanism was by measuring sound. However, this approach was not feasible, as normal operation of the adaptive mechanisms involves movements that generate significant noise. This would make it difficult to distinguish from sounds associated with actual damage. Instead, sound may be used as a supplementary indicator to identify deviations from normal operation. As a result, the test rig is not designed to be completely silent, but to dampen noise while still allowing changes in sounds to be perceived.

Additionally, methods for measuring misalignments were also discussed. Accurate misalignment measurement typically requires additional equipment, such as laser alignment systems or displacement sensors. Since those types of equipment rely on a fixed reference point in space, implementing such a setup would require extra supporting structures and precise calibration to maintain a stable reference. Although, due to physical space limitations and the complexity of integrating misalignment equipment, it was not considered feasible within the scope of this project.

9 Verification

The stability of the test rig was validated by conducting solid mechanics calculations and a modal analysis on the critical parts of the rig. No predefined requirements were initially established; instead, it was evaluated whether the results remained within acceptable engineering assumptions and engineering judgement.

9.1 Frequency Of The Rig

One of the main concerns for the test rig is the level of vibrations generated by impacts and component motion. In particular, there is a risk of resonance occurring if the excitation frequency coincides with the natural frequencies of the structure. Resonance may occur when the excitation frequency or its harmonics approach the natural frequencies of the system. To evaluate this risk, the natural frequencies of the most critical components were analysed. A modal analysis was therefore conducted to determine the natural frequencies of selected parts of the test rig.

The analysis began by calculating the excitation frequency f [Hz] based on the rotational speed of the drive unit, according to Equation (1). Where n is the rotation speed of the drive unit in revolutions per minute [rpm]. For further calculations, the rotational speed was converted into angular velocity ω [rad/s] using Equation (2).

$$f = \frac{n}{60} \quad (1)$$

$$\omega = \frac{2\pi n}{60} \quad (2)$$

To estimate the forces that could cause imbalance in the test rig, the centrifugal force was calculated using Equation (3):

$$F = mew^2 \quad (3)$$

where m [kg] is the rotating mass, which is the mock impeller including the adaptive mechanism, e [m] is eccentricity, defined as the distance between the centre of the mass and the geometric centre, and ω [rad/s] is the angular velocity. Since the mock impeller is mounted without precise alignment features, two eccentricity cases were considered: a design and worst-case scenario. These resulted in calculated forces of 25N (design case) and 62N (worst case).

To determine which component would be dimensioning for the test rig, corresponding calculations were also performed for the linear actuator. The motion of the actuator was approximated as sinusoidal.

The amplitude A [m] was calculated using Equation (4) and the linear speed v [m/s] of the linear actuator was calculated using Equation (5). Where the lead represents the travel distance per revolution of the roller screw, and RPM is its rotational speed.

$$A = \frac{stroke}{2} \quad (4)$$

$$v = \frac{lead \cdot RPM}{60} \quad (5)$$

Assuming periodic motion, the frequency f [Hz] was estimated using Equation (6):

$$f = \frac{v}{4A} \quad (6)$$

This relation is based on the actuator traveling a total distance of $4A$ during one full cycle. The resulting dynamic force generated by the actuator was then calculated using Equation (7). Which corresponds to the inertial force of a body undergoing harmonic motion. The calculated force was 15N.

$$F = mA(2\pi f)^2 \quad (7)$$

Since the drive unit generated higher forces compared to the linear actuator, it was identified as the dimensioning case for the design of the test rig components. Based on this, the natural frequencies of the relevant parts were evaluated.

The natural frequency of the vertical plate supporting the drive unit was calculated using formula from KTH Solid Mechanics Handbook (Sundström, n.d., p.160). The plate was assumed to be clamped on all edges and modelled as a rectangular bending plate.

Material properties for aluminium were used: Young's modulus E = 70 GPa, density ϱ = 2700kg/m^3, and Poisson's ratio v = 0.33. The plate dimensions were defined as a = 0.49m (height), b = 0.615m (width), and h = 0.025m (thickness). The plate stiffness was calculated using Equation (8):

$$D = \frac{Eh^3}{12(1-v^2)} \quad (8)$$

where E is Young's modulus, h is the thickness, and v is Poisson's ratio. The resulting stiffness was calculated to be 102,284 Nm. Using Equations (8), (9) and (10), the natural frequency was determined. The coefficient λ = 35.99 was used, corresponding to a clamped rectangular plate configuration:

$$\omega = \frac{\lambda}{a^2}\sqrt{\frac{D}{\rho h}} \quad (9)$$

$$f = \frac{\omega}{2\pi} \quad (10)$$

The calculated natural frequency of the plate was 929Hz. However, the presence of a large hole in the plate to accommodate the drive unit reduces its stiffness. To account for this, a reduction factor of 20-40% was applied, resulting in an estimated frequency range of approximately 400-700Hz. When compared to the excitation frequency of the drive unit (50Hz), the natural frequency of the plate is significantly higher, indicating a low risk of resonance. Even with considering harmonics of the excitation frequency (e.g., 100Hz, 150 Hz), these remain well below the estimated natural frequency range and are therefore unlikely to induce resonance.

The natural frequency of the triangular supports was also evaluated, as these components are subjected to loading and may contribute to structural vibrations. The supports are made of aluminium with dimensions a = 0.42 m, b = 0.35 m, and h = 0.03 m.

For the analysis, the triangular geometry was approximated as a rectangular bending plate. Using Equation (8), (9), and (10), the natural frequency was initially calculated to be 1516 Hz. Although, the actual geometry includes cutouts and deviates from a full plate, resulting in reduced stiffness. To account for this, a reduction factor was applied. A factor of 0.6 was selected, as it represents a typical empirical estimate for plates with aggressive geometry reduction. This resulted in an adjusted natural frequency of approximately 910 Hz. Since this value is significantly higher than the excitation frequency range of 50-100 Hz. The triangular supports are not expected to experience resonance during operation.

The natural frequency of the linear guide rails was also calculated. The longest linear guide rails were selected, as they are expected to have lower stiffness and therefore a greater tendency to vibrate compared to shorter

rails. Therefore, the rails parallel to the linear actuator were considered, with a weight of 2.9 kg obtained from Creo. The rail was modelled as a straight beam, with its cross-section approximated as rectangular. The calculations were based on straight beam formulas from the Handbook (Sundström, n.d., p.154). The material was steel, with properties E = 206 GPa and ϱ = 7850kg/m^3. The dimensions of the rail were b = 0.028 m, h = 0.028 m, and L = 0.602 m. The second moment of area I [m^4] for a rectangular cross-section was calculated using Equation (11):

$$I_y = \frac{bh^3}{12} \tag{11}$$

Using Equations (11), (12), and (10), the natural frequency of the linear guide rail was determined. The coefficient μL = 4.730 was used, corresponding to the assumed boundary conditions of the beam:

$$\omega = (\mu L)^2 \sqrt{\frac{EI}{mL^4}} \tag{12}$$

The natural frequency of the rails was calculated to be 492 Hz. This is out of 50-100 Hz range, which indicates that resonance is unlikely to occur. The reason for evaluating the natural frequency of the components is to identify potential resonance, which would lead to increased vibration amplitudes and energy absorption within the structure. Overall, none of the critical components have natural frequencies within the same range as the excitation frequencies of the drive unit, and therefore resonance is not expected.

9.2 Deflection Of Tabletop And Guide Rails

The tabletop and the linear guide rail system was analysed to determine its deflection under the applied loads. The analysis was divided into two parts: deflection of the tabletop supporting the test rig and deflection of the linear guide rails themselves. The tabletop was evaluated firstly to determine if it would deform under the weight of the assembled components. An elementary beam bending case from the KTH Solid Mechanics Handbook (Sundström, n.d., p.357) was applied, as illustrated in Figure 28.

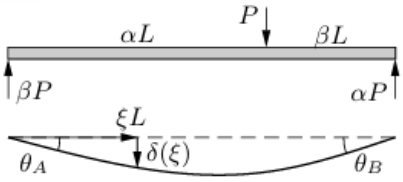


Figure 28 Elementary beam bending case

The vertical deflection δ [m] was calculated using Equation (13):

$$\delta_1 = \frac{Pl^3}{3EI}\alpha^2\beta^2 \tag{13}$$

where P is the applied load [N], l is the length of the tabletop [m], E is Young's modulus of steel [GPa], I is the second moment of area of the tabletop cross-section [m^4]. The parameters a and β are dimensionless coefficient accounting for geometric and loading conditions. The calculated deflection was 1.9×10^{-3}m, indicating that the tabletop remains sufficiently stiff under the applied load.

To further evaluate the deformation of the tabletop under operational loads, a finite element analysis (FEA) was conducted using Creo Ansys. The applied loads correspond to the force generated by the drive unit and the linear actuator, as initially calculated. As shown in Figure 29, the displacement in the direction parallel to the tabletop was approximately 0.17×10^{-3}m. This deformation is negligible, indicating that the tabletop provides sufficient stiffness to withstand the applied loads during operation. Moreover, the results are consistent with the analytical calculations, confirming that the tabletop deformation is minimal.

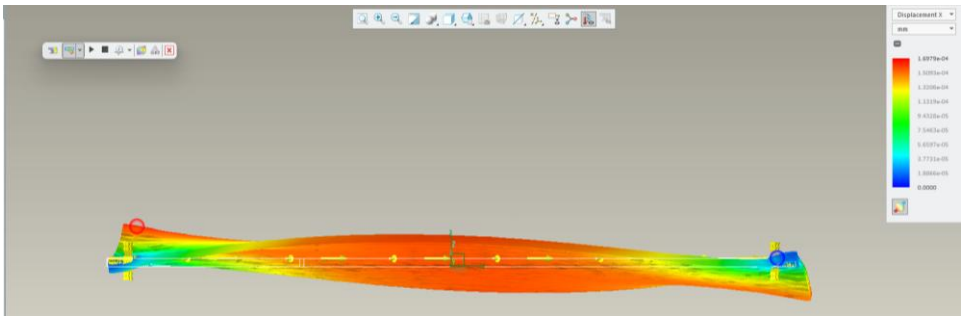


Figure 29 Displacement of tabletop under applied forces from Creo Ansys

The linear actuator plate was analysed to determine its deflection under the applied loads. The analysis combined two separate elementary beam bending cases. The first case considered the deflection caused by the weight of linear actuator. While the second case included the deflection caused by the moment generated when the actuator applies force on the mock impeller, creating misalignment. The dynamic loading was simplified as static loading, since the calculation was used as an initial estimate for the conceptual design.

The actuator weight was assumed to act at the centre of the plate. The vertical deflection was calculated using Equation (13), resulting in 4.79×10^{-6} m. The second elementary beam bending case considered the moment caused by the pressing force from the actuator, as illustrated in Figure 30.

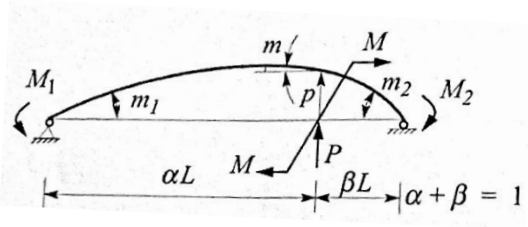


Figure 30 Elementary beam bending case

The deflection cause by the moment was calculated using Equation (14):

$$\delta_2 = \frac{Ml^2}{6EI}\alpha\beta(1-\beta) \qquad 14$$

where M [Nm] is the bending moment caused by the actuator force. The calculated deflection was -19.01×10^{-6}m. The negative sign indicates the direction of deflection. The total deflection was calculated by adding the two deflections, which resulted in -14.21×10^{-6} m. This was considered very small bending of the plate where actuator is mounted on, which indicates that the plate remains stiff during applied loads.

In addition, the deflection of the linear guide rail parallel to the linear actuator was analysed (see Figure 31 for the structure). It was evaluated whether the rails will deform under the weight of the components and moment caused by the force of the actuator. The corresponding process was used, combining the two beam bending cases with Equation (13) and (14). This resulted in the total deflection of the linear guide rail to be -39.6×10^{-6}m.

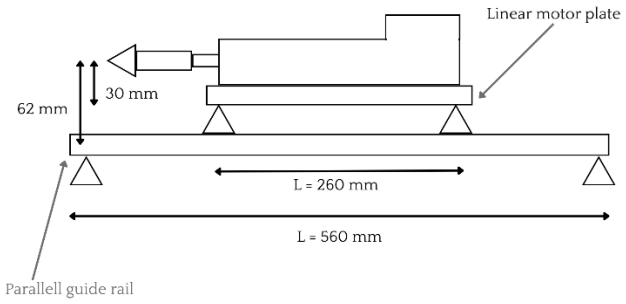


Figure 31 Simplified illustration of linear actuator on linear guide rail, front view

Moreover, the deflection of the linear guide rails perpendicular to the linear actuator was also calculated, illustrated in Figure 32. It was assumed that the linear actuators position was aligned with the mock impellers eye. It was evaluated if the rail would deform under the weight of the assembled components. Therefore, Equation (13) was used, which resulted in 28×10^{-6}m. As a result, both the parallel and perpendicular linear guide rails are stiff enough under the applied loads.

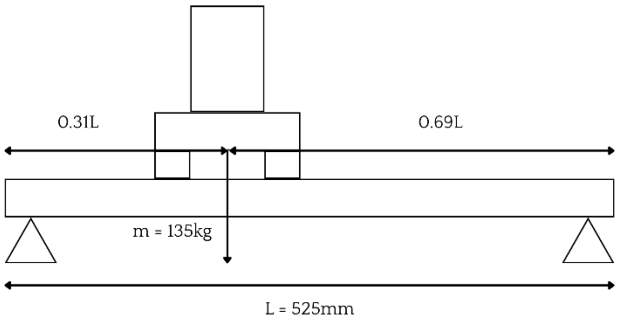


Figure 32 Simplified illustration linear actuator on linear rail, right view

9.3 The Force Of Bolts

Another concern for the test rig was that the screws and bolts may fail or loosen due to vibrations during operation. To evaluate this, calculations were performed to assess the risk of bolt failure and the potential for self-loosening. The initial objective was to determine whether the bolts could fail under applied loads. The bolts analysed were the ones securing the vertical plate, as this component carries the drive unit and is therefore subjected to the highest loads. The plate is fastened using twelve M10 bolts on the front and four M8 bolts on the side. It was assumed that the load is equally shared among the bolts in each group. The

force generated by the rotating mock impeller was taken 25-62 N. The force per bolt due to rotation was calculated using Equation (15):

$$F_{screw} = \frac{F_{rotating\ force}}{number\ of\ screws} \quad 15$$

This resulted in a force range of 2.1-5.2 N per M10 bolt. For the M8 bolts, the corresponding force was calculated as 6.3-15.5 N per bolt.

The force contribution from the linear actuator was also considered. The axial force per bolt was calculated using Equation (16):

$$F_{axial} = \frac{700}{number\ of\ screws} \quad 16$$

This resulted in approximately 58.3 N per M10 bolt and 175.7 N per M8 bolt. The combined loading was calculated using Equation (17):

$$F_{res} = \sqrt{{F_{screw}}^2 + {F_{axial}}^2} \quad 17$$

This resulted in total forces of approximately 58.4 N per M10 bolts and 175.7 N per M8 bolt, showing that the axial load dominates the combined loading. The calculated forces per bolt are small compared with the load carrying capacity of standard M8 and M10 bolts. Therefore, bolt failure due to direct loading is not expected.

To evaluate the risk of bolt loosening due to vibration, a slip-based check was performed using the worst-case force from the rotating mock impeller, F = 62 N. The required preload was calculated using Equation (18):

$$\sum F_p \geq \frac{F}{\mu} \quad 18$$

where F_p is the preload force and μ is the friction coefficient between steel and aluminium, since the bolts will be made of steel and vertical plate will be of aluminium. Using these given values, the total preload was calculated as 413.3 N. This corresponds to approximately 34.4 N per M10 bolt and 103.3 N per M8 bolt. These required preloads are far below the preload normally achieved in properly tightened steel bolts, and the risk of self-loosening due to vibration loads is therefore considered to be low.

10 Discussion

The following chapter includes evaluations of the research questions, the applied research methodology and future work with the test rig.

10.1 Main Findings And Interpretations

This thesis has explored a conceptual design for a test rig aiming to reproducing and analysing damage in the adaptive mechanism under controlled conditions. The problem statement identified three main challenges within the adaptive mechanism: mechanical axial collision, misalignment, and leakage. This thesis focuses on mechanical axial collision and misalignment. Leakage was excluded because Xylem already has an existing method for testing leakage-related issues. Hydraulics forces were also excluded, as the use of water would introduce unpredictable variables and reduce the repeatability of the test. The RMTR will apply a radial or shear force on the impeller, to recreate how the object collides with the impeller in different positions.

The first research question concerned which parameters are most relevant for identifying damage in the adaptive mechanism. The selected parameters were vibration, temperature, current consumption, time and number of impulses. Vibration is measured using an accelerometer on the drive unit shaft and is relevant, since sudden increase may indicate a fault in either the adaptive mechanism or the test rig. Temperature is measured using a thermocouple and can indicate excessive friction or malfunction in components such as the lathe centre, roller bearing or linear actuator. Current consumption from the drive unit is valuable, as an increase may show that a component is lodged or that the adaptive mechanism is no longer moving as intended. Finally, the number of impulses and time is counted by the linear actuator, which allows damage progression to be linked to repeated loading cycles and its time occurrence.

Together with this, the selected parameter does not automatically identify which component has failed, but they provide an indication of when abnormal behaviour occurs. The adaptive mechanism must still be removed and inspected manually after testing to determine the exact damage. However, the measured data can show at what time, after how many impulses, or under which loading conditions the behaviour changed. This allows different prototypes to be compared under the same test conditions.

The second research question concerned which motions the test rig should simulate to reproduce the damage in the mechanism. The concept enables two main types of tests: one for repeated axial loading and one for misalignment. Axial movement is expected to affect components such as the spring and sleeve, while misalignment may affect the spline teeth, spring and inlet. Rotation was also included, as the mechanism is exposed to rotational motion during real conditions, and this may contribute to wear in the spline teeth. Therefore, the rig should simulate repeated axial movement, rotation and adjustable misalignment.

A key design decision was whether the rig should apply a colliding force or a pushing force. Initially, a colliding force was considered because it was expected to create severe damage in the mechanism. However, this solution was disregarded, as repeated impacts could damage the lathe centre before the adaptive mechanism itself fails. For each collision, the lathe centre would need to accelerate back to 3000 RPM, which could create an effect compared to a grinding machine. This would make the test unreliable, since the rig could fail before achieving its purpose. Therefore, a pushing force was selected instead. Although, this does not fully represent a real collision, the return movement of the mock impeller to its distal rest position may still create the similar effect in the mechanism.

The third research question addressed how the test rig can be designed to minimise manual labour during the test. In comparison with the CTR, the RMTR is less labour intensive and time consuming. The existing method in the CTR requires a whole working day to achieve certain results. Fabrics are physically thrown into the wastewater pump repetitively, while in comparison the RMTR does not require any physical labour during operation. The test rig needs to be assembled and have chosen input data, which is a similar process that the CTR requires. For the RMTR the user does not need to be present while the test is running. The test rig can be left unattended since it is a controlled environment and it has its own emergency measures.

A key objective was to accelerate damage by applying repeated, controlled loading to the adaptive mechanism. Instead of waiting for damage to occur during normal pump operation, the rig exposes the mechanism to repeated axial movement, rotation and misalignment in a shorter time. The test setup allows adjustment of force, rotational speed, and misalignment position, as well as increased test severity when required. However, the results should be used for comparison between prototypes rather than as a direct prediction of service life, since the test conditions differ from real pump operation. When the same test procedure is used for different prototypes, the number of impulses and the measure behaviour can be compared to evaluate which design is more resistant to damage. Over time, it may also be possible to investigate whether there is a relationship between number of impulses in the RMTR and the number of fabrics used in the current CTR method.

The requirement specification was mainly used as a guide for the conceptual design, and the concept fulfils several of the main functional requirements. These include repeated axial loading, rotational motion, adjustable misalignment, relevant measurement parameter and reduced manual labour. The concept also considers substantial aspects such as adjustable test conditions by having the linear actuator having two modes, and adaptability by enabling movement of the linear actuator.

Overall, the RMTR addresses the research questions by identifying relevant damage parameters and defining the required motions. Moreover, showing how repeated controlled loading can accelerate damage, and reducing manual labour compared with the current method. However, since the rig is conceptual, several requirements must be verified through further development and physical testing.

The verification presented in Chapter 9 indicates that the test rig is highly feasible to construct, as the calculated values were considered acceptable for the forces and loads to which the system will be exposed. Although no predetermined target values were established at the beginning of the project, the results obtained demonstrate that the design is sufficiently robust and suitable for practical implementation.

This project contributes to the research field through the investigation of a phenomenon that has not previously been studied: the combination of rotational and linear motion within the same mechanical body. Moreover, the developed test rig provides Xylem with a valuable tool for continued product development, enabling more efficient fault analysis, evaluation, and optimisation of future prototypes.

10.2 Evaluation Of The Research Methodology

The selected methods were considered suitable for the thesis since this thesis was a development project, which required continuous iteration and improvement. There was no prior work or research conducted before this thesis was began; therefore, the information had to be developed independently. The double diamond framework supported both broad exploration and systematic narrowing of concepts, which was beneficial due to the initially wide project scope and limited prior research within the area.

Semi-structured interviews provided valuable first-hand insight into user needs and existing test rig challenges, while white and grey literature enabled inspiration to be gathered from other industries. FMEA and FTA were effective tools for identifying relevant failure mechanisms and determining which motions the RMTR should replicate.

Concept generation methods encouraged broad exploration of design solutions, whereas the criteria weighting method allowed for system evaluation of concepts against the requirements specification. Calculations, simulations, and simplified prototype testing were additionally substantial for assessing structural feasibility and operation behaviour.

Alternative approaches, such as simulation-driven development and stage-gate models, were considered. However, these approaches were considered less suitable since the project required flexibility and continuous feedback through the development process.

10.3 Limitations In Work Performed

The scope of the project was to develop a conceptual design; therefore, it is not fully optimised for direct implementation. Design choices were made to simplify development and enable testing within the scope of the project. The prototype developed to test the concept was a simplified version of the real test, with some components and motions having to be altered. This might have influenced the accuracy compared to how a real version of the concept would act. The forces and rotational speed had to be lowered drastically in comparison to the intended forces for the RMTR. The prototype test did also not aim at recreating long term wear effects on the components within the adaptive mechanism, or even evaluated any damage done to the adaptive mechanism.

10.4 Future Work

Since the RMTR is conceptual, further work is required before a physical prototype can be manufactured and validated. The following points are recommended for future development:

- The recommended linear actuator should be ordered and tested together with a mock impeller manufactured in grey iron. This would allow the interaction between the actuator, the mock impeller and the adaptive mechanism to be evaluated under more realistic conditions.
- The spring should be evaluated to determine whether it is at risk of failure under the load applied by the linear actuator.
- The RMTR should be placed on a solid and flat surface, preferably closer to the ground. The goal would be to improve the stability of during testing by reducing the risk of tilting.
- The emergency stop system should be developed, to stop both the linear actuator and the drive unit simultaneously. An automatic stop function could also be added, for instance, when the safety barrier is lifted during opening. In addition, an alarm could be included to warn the user if critical temperature or current levels are reached.
- The user interface and measurement equipment should be assembled next. The system should be able to store collected data and display the selected parameters.
- Once a physical prototype has been built, the noise level should be measured to determine whether additional sound dampening is required in the safety barrier. The noise level should not exceed the recommendation in the requirement specification.
- A reference test should be developed to determine how long each test should run and what loading conditions should be used.

- The concept could be further developed, enabling use for additional impeller models besides the 6020 and 6030. This would increase the usefulness of the rig within future development projects at Xylem.
- If several impellers or adaptive mechanisms need to be tested simultaneously, the concept should be developed into multiple test rigs. This would increase testing capacity and reduce the total time required for prototype evaluation.

11 Conclusion

The objective of this project was to describe the development of a conceptual test rig for the adaptive mechanism within Xylem's Concertor wastewater pumps. The test rig was named RMTR and enables the R&D departments to understand how different motions on the adaptive mechanism causes different damage. The final concept offers the department a structured tool for identifying where failure occurs within the adaptive mechanism.

A concept has been designed and a simplified version in the correct scale has been tested. Material choices and structural design were evaluated to ensure durability under the applied loads. The rig allows testing of linear loads and misalignment on the adaptive mechanism within the 6020 and 6030 model of Xylem impellers. When conducting the linear test, the linear actuator will be equipped with a lathe centre, enabling rotation of the mock impeller without inducing rotation of the actuator shaft. The linear actuator will be moved along the linear guide rails until the lathe centre is aligned with the mock impellers eye. During the misalignment test the attached part on the linear actuator will be a roller bearing. The linear actuator will be moved on the linear guide rails to be near the circumference of the mock impeller, for inducing the greatest misalignment. When the attachable part or the mock impeller is changed between tests the linear actuator can be moved on the linear guide rails to create space for assembly. After the linear actuator has been moved, clamps will be secured to stop any unwanted movement during the test.

In conclusion, the projects objectives have been achieved by developing a conceptual test rig that meets the main needs identified by Xylem. The concept provides a structured and repeatable process to investigate damage in the adaptive mechanism, while also reducing manual labour. Although further development and prototype testing is required, the concept fulfils its intended purpose and provides a strong foundation for future development.

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Appendix A

The following is the final version of the requirement specification for the test rig, which have been re-developed during the project. The specification includes general requirements, size, sustainability, safety, regulatory and wishes.

Requirement specification

- Applying sufficient loads to cause failure in the adaptive mechanism
- Apply axial forces repetitively
- Allow for minimum 170 N axial force
- Allow for maximum 2000N axial force
- Apply rotational force
- Allow for maximum 3000RPM
- Allow for adjustment to the applied force
- Allow for adjustment of the rpm
- Measure number of impulses
- Allow for adjustment to the position misalignment is applied
- Allow for adjustment the degree of the misalignment
- Identifying faults, measure the following parameters
    - Vibrations
    - Effect
    - Temperature
- The system should be adaptable to different adaptive mechanisms (impeller sizes)
- Ability to change impeller without disassembly of rig
- Allow for use in the temperature interval of 0 to +65°C in the impeller
- Allow the user to set how long (hours) the rig will run
- Allow for a screen display
    - Timestamps
    - Time running
    - Expected time
    - On/off button
- Allow for data to be stored locally and online

Size

- Working height above knee height and under shoulder height
- Weight maximum 4 ton
- Limited size (120x80 cm)

Sustainability

- Allow for repair and recycle
- Use materials that are nontoxic
- Design constraint: minimise composite or bonded materials

Safety and regulatory

- Noise level shall not exceed 85db
- Vibration shall meet AS ISO 5349.1-2013
- Offer containment of impeller when test is running to ensure that access to the dangerous parts is prevented.
- The system must include an emergency safety stop
- The system must include indication of operating status (when test rig is running and not)

Wishes

- The rig should be easy to operate
- Automatic stops when malfunctions arise
- The test can run automatically during the test cycle
- Allow for several impellers to be tested simultaneously

Appendix B

The following shows the functional analysis for the test rig. It was a fundamental part of the concept generating process as it guided which components were necessary to carry out each function.

Function analysis for the test rig

Function	Type of function	Class	Limitation
Damage mechanism within accelerated time	MF (main function)	N (necessary)	
Test the mechanism	MF (main function)	W (Wished)	
Enable switching the machine on and off	SF (Sub- function)	N	
Provide a timer	SPF (Support function)	W	
Enable time measurement	SF	N	
Offer a relief/support surface	SF	W	
Facilitate cleaning	SF	W	
Minimise tipping risk	SF	N	
Provide mounting of the machine	SF	N	
Provide feedback	SF	W	
Simple handling of the rig (easy to use)	SF	N	
Offer automatic operation	SF	W	
Enable troubleshooting of the rig	SF	N	
Warning system in case of faults in the rig	SPF	N	
Provide emergency stop when the lid is open	SPF	W	
Minimise pinch risk	SF	N	
Minimise risk of cuts	SF	N	
Minimise risk of crushed body parts	SF	N	
Prevent loose/broken components from flying off	SF	N	
Not break from the forces / Minimise wear	SF	N	
Provide sound dampening	SPF	N	If exceeding 85 decibel levels
Enable long service life (rig)	SF	W	
Enable visual monitoring	SF	W	Transparency

Read measurement values	SPF	N	
Log the measurement values	SPF	N	
Inform when components break	SPF	W	
Enable disassembly and inspection of the adapter mechanism	SF	N	
Enable testing of different sizes	SF	N	Decreases durability
Enable test of impellers simultaneously	SF	W	
Simulate forces	SF	N	
Provide rotation of components*	SF	N	
Provide adjustment of rotational speed*	SF	N	
Provide axial movement (spring)	SF	N	
Provide adjustment of axial speed	SF	N	

***** One of the core components of the test rig is the motor, since the movement of the adaptive mechanism has to be both axial and rotational simultaneously. A motor that was analysed is the linear motor, which works with magnetic fields. Another was the piston and the last is pneumatic cylinder. The different components were compared to each other with speed in mind. However, one of the factors that were crucial was that the movement would not be determined by an exact distance. Since when the test rig is operating, the components of the adaptive mechanism may be damaged and the distance the axial movement would become more. Therefore, it was important that the movement happens until the component makes contact with something, to thereafter move back again.

Appendix C

This Appendix introduces the criteria weighting method for the evaluation of the three concepts. Criteria weighting method was used with an early draft of the requirement specification. Therefore, some of the requirements are not relevant in the final concept.

Criteria weighting method

Criteria	Weight	Concept 1: Dual motor		Concept 2: Motor on rail		Concept 3: High pressurized water	
		Score	Total	Score	Total	Score	Total
Applying sufficient loads to cause failure in the adaptive mechanism	5	3	15	4	20	4	20
Apply axial forces repetitively	5	2	10	4	20	3	15
Allow for minimum 800 N axial force	5	5	25	5	25	5	25
Allow for maximum 5000N axial force	5	1	5	5	25	5	25
Apply torque to the motor	5	4	20	5	25	0	0
Apply rotational force	3	4	12	5		5	
Allow for minimum 1000 rpm	3	5	15	5	15	5	15
Allow for maximum 4000 rpm	3	1	3	5		5	
Simulate the real conditions of the impeller	5	1	5	1	5	1	5
Allow for adjustment to the forces applied	5	5	25	5	25	5	25
Measure number of impulses	5	5	25	5	25	4	20
Allow angle of misalignment	5	5	25	5	25	5	25
Measure for identifying faults			0		0		0
Vibrations	1	3	3	3	3	2	2
Effect	1	3	3	3	3	3	3
Sound	1	0	0	0	0	0	0
Distance	1	0	0	3	3	3	3

	The system should be adaptable to different adaptive mechanisms (pump sizes)	4	4	16	4	16	4	16
	Ability to change impeller without disassembly of rig	5	5	25	4	20	4	20
	Allow for use in the temperature interval 0 to +65°C in the impeller	2	5	10	5	10	5	10
	Allow the user to set how long time the rig will run	5	5	25	5	25	5	25
	Allow for a screen display	2	5	10	5	10	5	10
								0
	Expected time to finished test cycle	4	5	20	5	20	5	20
	On/off button	5	5	25	5	25	5	25
	Allow for data to be stored locally and online	1		0		0		0
	Working height above knee height and under shoulder height	4	5	20	5	20	5	20
	Test rig weight of maximum 4ton	2		0		0		0
	Limited size (120x80 cm)	3	5	15	4	12	4	12
	Allow for repair and recycle	3	3	9	3	9	3	9
	Use materials that are nontoxic	3	4	12	4	12	4	12
	Noise level shall not exceed 85db (how do we measure that)	4	2	8	2	8	2	8
	Vibration shall meet AS ISO 5349.1-2013	4	1	4	1	4	1	4
	Offer containment that can withstand impact loads of broken components	5	5	25	3	15	3	15
	The system must include an emergency safety stop	4		0		0		0
	The system must include indication of operating status	2		0		0		0
	Wishes							
	The rig should be easy to operate	4	5	20	3	12	4	16

	Automatic stops when malfunctions arise	1	1	1	1	1	1	1
	The test can run automatically during the test cycle	4	5	20	5	20	5	20
	The system should notify the user to damage in the adaptive mechanism	1	1	1	1	1	1	1
	Allow for several impellers to be tested simultaneously	3	3	9	3	9	3	9
	Design constraint: Minimise composite or bonded materials	3	1	3	1	3	1	3
Total		**137**		469		471		439

Appendix D

This appendix illustrates the method used to compare the various mock impellers that was created. The criteria weighting method was applied to compare the designs against the key requirements of safety and weight.

Criteria weighting method

		Turbine blades		Turbin blades small		Biconcave		Biconcave and blades		Rounded corners		Curved blades	
	Criteria weight	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total
Weight	5	1	5	5	25	2	10	4	20	1	5	5	25
Safety	4	3	12	2	8	4	16	3	12	4	16	2	8
Total			17		33		26		32		21		33

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